

Low emissions analysis platform model for renewable energy: Community-scale case studies in Nigeria

B. Ugwoke^{a,b,*}, S.P. Corngati^a, P. Leone^a, R. Borchellini^a, J.M. Pearce^{b,c}

^a Energy Center Lab, Department of Energy Politecnico di Torino, Italy

^b Department of Materials Science & Engineering and Department of Electrical & Computer Engineering, Michigan Technological University, Houghton, MI, 49931, USA

^c Department of Electronics and Nanoengineering, School of Electrical Engineering, Aalto University, Espoo, Finland

ARTICLE INFO

Keywords:

Community-scale energy planning
Clean energy analysis
Energy access and productive services
Energy system model
Sustainable development
Renewable energy

ABSTRACT

Community-scale energy planning entails concerted efforts for realizing localized energy and emission plans geared towards achieving energy sustainable communities. This is largely under-investigated for the developing world together with exploring the synergies between community planning and energy planning strategies. This study performs community-scale energy planning from 2015 to 2040 for Onyen-okpon and Giere, two rural communities in Nigeria. Utilizing the hybrid integration approach with the Low Emissions Analysis Platform model, the study investigates the implications for transforming rural communities to sustainable renewable based energy supply and quantitatively analyses the strategy integration based on four scenarios: the reference scenario and three composite demand side management scenarios. The results show that the integration of strategies has clear benefits with the composite scenarios availing reduced energy use with significant reductions in energy demand and greenhouse gases emissions. The third demand side management scenario provides the best performance among the composite scenarios in terms of demand and emissions reductions, energy conservation, and overall reduction in energy expenditures. The integration approach provides useful insights for design and implementation of localised energy policies across different sectors in the community where productive uses are taken into account for clean and sustainable economic production. This avails the opportunity to concurrently define and reach multiple targets, goals and impacts in reducing emissions while ensuring energy access and productive services.

1. Introduction

Energy access is crucial for economic development (Rocco, Tonini, Fumagalli, & Colombo, 2020) and the level of energy access is seen as a measure of the level of development achieved by a country (Toman & Jemelkova, 2003), region (Sokona, Mulugetta, & Gujba, 2012) or locality (Kanagawa & Nakata, 2008). Developing countries, especially those in Sub-Saharan Africa are generally plagued with low energy access (Bertok & Bartos, 2020). This is manifested in their energy intensities and per capita consumption of energy (Voigt, De Cian, Schymura, & Verdolini, 2014), parameters reflective of the development level and economic disposition in the region (Chen, Huang, & Zheng, 2019).

Within these countries and at local scales, the rural areas have severely underdeveloped energy system infrastructures (Ali, Chiu, Aghaloo, Nahian, & Ma, 2020). A global consensus has advocated for

uptake of decentralized renewables as competitive options to improve rural energy access (Deichmann, Meisner, Murray, & Wheeler, 2011), including deploying decentralized mini grids (Szabó, Moner-Girona, Kougias, Bailis, & Bódis, 2016), decentralized energy networks (Alstone, Gershenson, & Kammen, 2015), and solar energy home systems (Yadav, Davies, & Sarkodie, 2019) and home micro grids (Marzband, Alavi, Ghazimirsaeid, Uppal, & Fernando, 2017). This thought process is reiterated by the integrated rural energy planning (IREP) framework which is poised to change the narrative for rural areas and spur sustainable development when diligently followed (Ugwoke, Gershon, Becchio, Corngati, & Leone, 2020). This framework buttresses community scale energy planning (C-SEP) (Karunathilake, Perera, Ruparathna, Hewage, & Sadiq, 2018).

C-SEP has been well researched for the developed world spanning studies on urban energy planning approaches (Torabi Moghadam, Delmastro, Corngati, & Lombardi, 2017), transition strategies for post

* Corresponding author at: Energy Center Lab, Department of Energy Politecnico di Torino, Italy.

E-mail address: blensing.ugwoke@polito.it (B. Ugwoke).

carbon societies (Geng, Fujita, Bleischwitz, Chiu, & Sarkis, 2019), smart city energy platforms (Pardo-García et al., 2019), holistic planning frameworks for urban areas (de A. Collaço et al., 2019), pathways for transitioning to sustainable energy futures (Chen, Xiong, Li, Sun, & Yang, 2019) etc. These studies have highlighted the importance of C-SEP for driving energy demand and emissions reduction, climate resilient strategies and urban development planning.

Additionally, with C-SEP there is the possibility to combine energy planning (EP) strategies with community planning (CP) strategies to drive localized energy policy design and spur low-carbon development. The EP strategies encompass energy management practices and uptake of renewable energy (RE) generation technologies while the CP strategies encompass local resources management (de A. Collaço et al., 2019). However, very few studies have focused on the developing world and the peculiarities regarding C-SEP such that can spur improved energy access and invigorate local economic development. C-SEP is crucial for dealing with contemporary issues such as limited conventional energy resources (Wesseh & Lin, 2017) and the devastating effects of climate change (Zafar, Shahbaz, Hou, & Sinha, 2019), already been felt by rural communities that contribute the least to its proliferation (Edmond, 2019). It would provide opportunities for exploring both indigenous RE potential (Yilmaz Balaman & Selim, 2016) and incorporating demand side management (DSM) based on energy efficiency (EE) whilst developing the energy system infrastructure (Farzaneh, Doll, & Puppim De Oliveira, 2016).

From a developing country perspective, research endeavours would strive to identify an optimal mix of energy supply infrastructure for defined energy demand from a cross-sectoral system perspective and ultimately improve the local (rural) energy access situation. This would take cognisance of the influential role cultural dichotomies and local conditions play in determining this optimal mix (Power for All, 2017).

To address the identified research gap, this study illustrates C-SEP for the developing world by means of clean energy system analysis. Based on the hybrid integration (HI) approach (Indra Al Irsyad, Halog, Nepal, & Koesrindartoto, 2017), energy demand from bottom-up building models are incorporated into the Low Emissions Analysis Platform (LEAP) model (Heaps, 2020) to investigate the implications for transforming rural communities to sustainable RE based energy supply considering the locally-built environment and its affiliated energy sub-sectors. The study entails modelling the long-term energy demand and supply evolution from 2015 to 2040 across two rural communities in Nigeria where different energy end uses are included such as residential and productive energy services. The modelling is based on composite scenarios and optimization analysis for plausible generation technologies. This would facilitate strategic decision making to select the optimal mix of energy supply technologies, support effective policy formulation on the uptake of non-dispatchable renewables and rural development initiatives, support preliminary design studies for deploying sustainable hybrid energy systems and ultimately provide a reference for future C-SEP in the developing world.

The remaining structure of this paper is as follows. Section 2 outlines the literature review on energy system analysis and models. Section 3 describes the modelling approach used. Section 4 presents and discusses the results and policy implications. The conclusions are presented in Section 5.

2. Literature review (energy system analysis)

According to Jaccard (2006), an energy system is defined as the combination of processes for “acquiring and using energy in a given geographic scope or economy”. This definition is extended to include all physical entities within the geographic boundary and all the related traceable upstream flows (Keirstead, Jennings, & Sivakumar, 2012). Energy system analysis encompasses methods employed to investigate the underlying operating principles of the energy system and its components. It proffers a concise scheme to quantitatively examine the

plausible development path of future energy system from the technical and organizational aspects (Witt, Dumeier, & Geldermann, 2020).

An energy system model embodies a combination of complex relationships and processes from extracting primary energy resources through final energy consumption to supply services and goods (Witt et al., 2020). It allows for abstracting energy systems and analysing the complex relationships among the energy demand, energy supply, the economy and the environment (Bhattacharyya & Timilsina, 2010). Diverse energy system models have been developed, which consider one or multiple energy sectors, environmental related issues, energy supply security, costs, or environmental impacts of energy consumption and production. These models may be differentiated by their techniques, data and skill requirements, technology specification, scope and purpose, computing demand, etc. This diversity in turn informs many categorization schemes employed in existing literature.

2.1. Categorization schemes for existing energy system models

While there is no strict scheme for grouping energy system models, several attributes have been used for categorizing existing energy models. This lack of transparency and standardization in categorizing existing models has made it difficult to ascertain their suitability for answering major energy policy questions (Pfenninger et al., 2018), Fig. 1 provides the broad categorization of schemes encountered in literature based on different criteria including general and specific purposes ((Van Beeck, 1999); trans-disciplinary criteria (Sola, Corchero, Salom, & Sanmarti, 2020); geographic focus criteria (Dioha, 2017); multi-dimensional criteria (purpose, structure and mathematical description) (Twaha & Ramli, 2018); electric supply planning criteria (Rojas-Zerpa & Yusta, 2014); energy policy issues criteria (Savvidis et al., 2019)).

The analytical approach has emerged the most common categorization scheme used in literature. This is based on three modelling approaches: 1) top-down approach, 2) the bottom-up approach and the 3) hybrid approach. The top down approach is used for aggregated analysis as it is economic based and focusses on economic theory-related interactions and system feedback (Dioha, 2017). The bottom-up approach is engineering based and focusses on energy sector technologies and is usually adopted for disaggregated level analyses (Dioha & Emodi, 2019). The hybrid approach combines the strengths of both bottom-up and top-down approaches to resolve their apparent weaknesses. This is done by HI approach based on three integration strategies (Indra Al Irsyad et al., 2017): 1) bottom-up demand variables modified into endogenous variables (Murphy & Jaccard, 2011), 2) the top down input data disaggregated into several specific technologies (Dai, Masui, Matsuoka, & Fujimori, 2011) and 3) other independent models to provide input data for existing energy models (Espinoza, Bautista, Narváez, Alfaro, & Camargo, 2017).

There are several “ready-made” software packages based on energy system models which are described in (Van Beeck, 1999), however, there is a consensus that many of the existing models are better suited for developed countries than developing countries (Bhattacharyya & Timilsina, 2010).

2.2. Community scale energy planning (C-SEP)

C-SEP entails concerted efforts for realizing localized energy and emission plans geared towards achieving energy sustainable communities. This is usually centred around three initiatives; EE, energy conservation and development of RE based energy supply (Karunathilake et al., 2018). This planning approach buttresses distributed generation and considers the availability of viable indigenous energy resources in the community. It seeks to align the energy system with the local needs and prevalent opportunities such that can catalyse local stakeholders involvement to foster the deployment of localised energy systems and further the attainment of long-term energy independence and energy

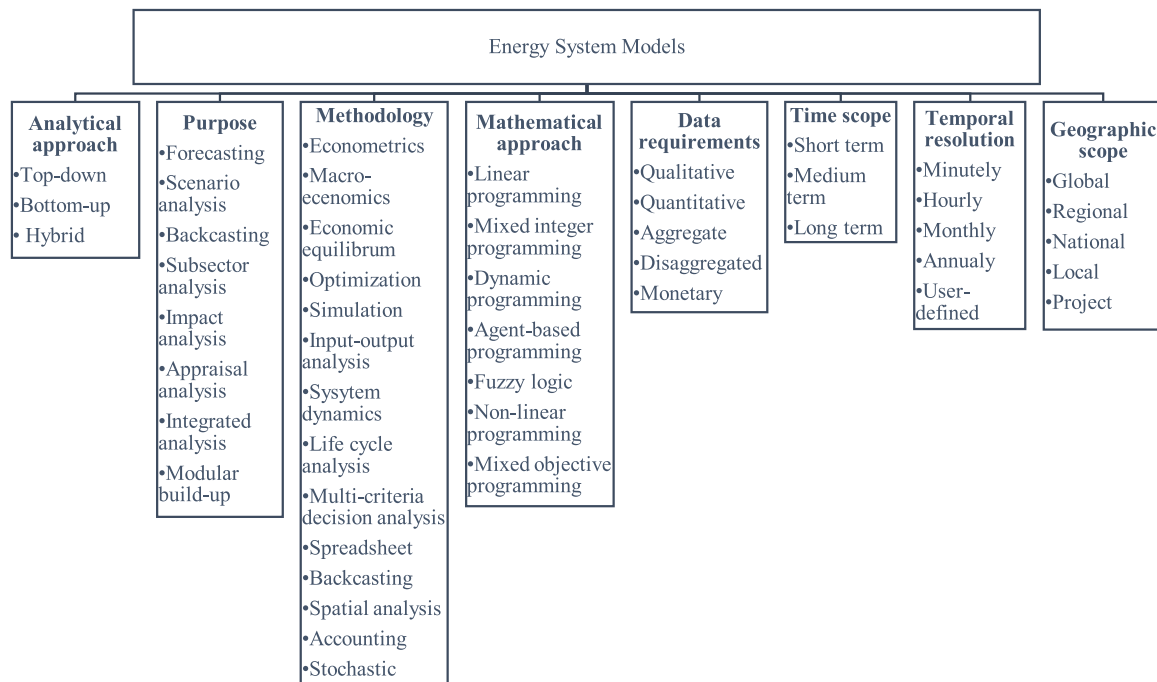


Fig. 1. Categorization schemes for energy system models.

security in the community (St. Denis & Parker, 2009).

In the developed world, this concept has been adopted for transitioning to sustainable urban energy systems and smart cities to enable improvements in resources efficiency, reductions in waste production and responsiveness to global climate change (Geng et al., 2019). The ‘Smart Urban Isle’ approach, a HI approach was adopted to develop innovative energy system configurations for Ramplaankwartier, a residential neighbourhood in Haarlem, the Netherlands (Jansen, Mohammedi, & Bokel, 2021). Witt et al. (2020) adopted a combination of scenario planning, energy system analysis, and multi-criteria decision analysis (MCDA) approaches to develop and evaluate energy scenarios for the power system in Lower Saxony, Germany. The scenarios development was facilitated by the morphological box approach, which provided a suitable tool for transparent energy scenario development (Witt, Stahlecker, & Geldermann, 2018). Additionally, a smart energy communities scenario calculator was developed and applied to Furuset, a suburb in Oslo, Norway. This availed a useful tool for providing decision support in defining development plans for C-SEP (Walnum et al., 2019). The MCDA approach was adopted to facilitate local EP for the Barreiro municipality in Portugal. This availed a decision support mechanism for the different stages of local sustainable EP processes from structuring end-use energy models to defining alternative energy action plans (Neves, Leal, & Lourenço, 2015).

Other authors adopted hybrid variant LEAP models for analysing urban energy systems in megacities including Beijing, China (Li et al., 2018) and Sao Paulo, Brazil (de A. Collaço et al., 2019). SureCity platform, an innovative EP tool for smart cities was designed by combining the integrated energy system model (ESM) and environmental assessment module (EAM) within the TIMES modelling framework (Pardo-García et al., 2019). Under the European integrative smart city planning (InSMART) project, an integrated planning framework was developed around four cities namely; Evora (Portugal), Nottingham (UK), Trikala (Greece) and Cesena (Italy) (Gargiulo et al., 2017). This framework used the TIMES model to derive city-ESM models which were integrated with the MCDA approach to foster comprehensive C-SEP and sustainable energy use geared towards the development of sustainable and resilient cities.

In the developing world, some studies adopted conventional energy

systems models for sector specific analysis. LEAP model was adopted for residential sector energy analysis in Dehli, India (Kadian, Dahiya, & Garg, 2007) while the MARKet ALlocation (MARKAL) framework was adopted for sectoral energy consumption studies in the Kathmandu Valley, Nepal (Shrestha & Rajbhandari, 2010). Howells, Alfstad, Victor, Goldstein, and Remme (2005) adopted the Integrated MARKAL-EFOM System (TIMES) for rural EP in Nkweletsheni, a South African village. Fuso Nerini, Dargaville, Howells, and Bazilian (2015) adopted the OSeMOSYS model for rural EP in Suro Craic, Timor Leste. The LEAP model was coupled with the MCDA approach to investigate and inform policy interventions for integrated energy and low-carbon development in Bangkok, Thailand (Phdungsilp, 2010). To the best of the authors knowledge, this is the only study that has adopted the HI approach at a local level in a developing country. This supports the view of the sparing use of this approach C-SEP in the developing world.

Very few studies have incorporated HI approach at country scale. Ali et al. (2020) adopted a hybrid MCDMA model to analyse the clean energy scheme in Bangladesh for different power generation technologies. The bottom-up agent-based strategy test-kit for electricity with renewables (BABSTER) model was developed and adopted for rural electrification planning in Liberia to complement the integrated resource planning (IRP) approach for electricity planning and facilitate engagement with stakeholders (Alfaro, Miller, Johnson, & Riolo, 2017). Rocco et al. (2020) combined the OSeMOSYS model with the Leontief Input-Output model to investigate electrification pathways in Tanzania. This approach is recommended for addressing the peculiarities of energy systems in the developing countries (Indra Al Irsyad et al., 2017) and advocated for performing C-SEP (Van Beeck, 1999). Therefore, this study goes a step further to demonstrate the application of this approach for C-SEP in the developing world. It strives to bring to the fore community perspectives and captures their local characteristics in a way that fuses CP with EP strategies into C-SEP.

3. Method and materials

3.1. The LEAP model

The LEAP model is ascribed as one of the most suitable energy system

modelling framework for developing countries (Dioha, 2017). LEAP is an integrated modelling tool supporting different modelling applications. On the demand side, it supports bottom-up/end-use accounting techniques and top-down/macroeconomic modelling. On the supply side, it supports accounting and simulation methodologies, as well as optimization modelling capabilities. LEAP can be used to create models of different energy systems across varying scales, each with its own unique data structures. LEAP model includes the Technology and Environmental Database (TED) that contains data on the costs, performance and emission factors for over 1000 energy technologies (Heaps, 2020). LEAP can perform comprehensive modelling of decentralized energy systems to inform rural electrification strategies (Pandey, 2002).

LEAP has the advantages of limited skill and initial data requirement as only the base year requires detailed statistical data (Bhattacharyya & Timilsina, 2010). LEAP is flexible and transparent such that allows for incorporating data and results from other independent models. This enables LEAP to act as a hybrid energy analytical tool. The time horizon of LEAP is unlimited and may be characterized by a series of years, split into varying resolutions of time slices (Emodi, Emodi, Murthy, & Emodi, 2017). LEAP is free for use in developing countries with a user-friendly interface (Heaps, 2020). Consequently, these characteristics make LEAP model most suitable for C-SEP and it was selected for use in this study.

3.2. Community scale energy planning: clean energy system analysis

The analyses performed encompassed long-range scenario analysis for energy demand, supply and emissions and integrated social cost-benefit analysis coupled with optimization analysis for plausible generation technologies. Based on the HI approach, the analyses adopted the demand estimated from bottom-up approaches (the reference building (RB) approach (Ugwoke, Becchio, Corgnati, Leone, & Pearce, 2020)) into endogenous demand in the community scale LEAP model. The emergent hybrid variant LEAP model was developed following some critical steps prescribed by Rogner (2017) as shown in Fig. 2.

The overall analyses sought to develop integrated clean energy strategies geared towards improving energy access at community scale. Two community-scale LEAP models were created for Onyen-okpon and Giere (see Fig. 3), rural communities with divergent climatic and geopolitical dispositions. Onyen-Okpon is a medium off-grid rural community in Obubra Local Government Area (LGA) of Cross-river state, South-East Nigeria while Giere is a small off-grid rural community in Dange Shuni LGA of Sokoto state, North-West Nigeria. They are both agrarian communities as agriculture accounts for the chief income generating activity. Local artisans whose business activities include provision of personal services (e.g. tailors, salons, retail trade, etc.) account for the commercial sector while community establishments (e.g. health centre and public schools) make up the community sector.

The categorization of the locally built environment informed the local characteristics of the communities including their energy use and apparent RE potential (quality and type of viable RE resource) (Ugwoke, Sulemanu, Corgnati, Leone, & Pearce, 2020). The perspectives and

priorities for the different stakeholders as conveyed in the choice and configuration of generation technology were based on the plans established by the rural electrification agency of Nigeria (REA) (Rural Electrification Agency (REA) (2017b)) and the composite scenarios reported in (Ugwoke, Becchio et al., 2020). The story and simulation approach which complemented scenario planning with energy system analysis was adopted for scenario planning (Weimer-Jehle et al., 2016).

The composite scenarios entailed a combination of strategies to drive local energy policy design based on the possible synergies between CP and EP to spur low-carbon development (Wang, Fang, Yin, & Chen, 2020). The CP strategies considered implementing advanced building retrofits based on the Nigerian Building Energy Efficiency Code (Geissler, Österreicher, & Macharm, 2018), adopting agricultural waste biomass for power generation aiming at better waste management and indoor environment quality (IEQ) criteria based on international standards on thermal comfort and indoor air quality (ASHRAE 55:2010, ISO 7730 and CEN EN 15251 (2007)). The EP strategies considered use of efficient lighting (natural and artificial) and equipment, harnessing local renewables (solar) for power generation and reducing power distribution losses. The ensuing holistic approach proffered strategies promulgating more sustainable C-SEP to derive optimistic results (including demand and emissions reductions, energy conservation, improved indoor environment and overall reduction in energy expenditures). The composite scenarios explored the opportunities for synergising EP and SP within the bounds of regional (rural) (Nduka, 2021) and national energy policy strategies (Nwozor, Oshewolo, Owioye, & Okidu, 2021).

The temporal scope (see Fig. 4) of the analysis period spanned 25 years from 2015 to 2040 to align with Nigeria's National Energy Master Plan (NEMP) (Energy Commission of Nigeria (ECN), 2014). 24 time slices were obtained based on seasonal and time of day variations in a year (Ezennaya, Isaac, Okolie, & Ezeanyim, 2014). The temporal scope captures these variations inherent in the annual demand which is in an hourly scale.

The LEAP model data structure as shown in Fig. 5, takes into account the total energy demand and transformation sectors, primary resources (mined natural resources and renewables) and secondary resources (produced secondary fuels) estimated in these communities. These are further explained in following sections.

3.2.1. The energy demand analysis

The energy demand analysis entailed disaggregated end-use analysis and scenario analysis. The total energy demand (ED) was calculated as follows (Chen et al., 2020);

$$ED_k = \sum_i \sum_j EI_{i,j,k} * AL_{i,j,k} \quad (1)$$

where AL is the activity level, EI is the energy intensity (energy use per unit of activity) for each sector i , j is the device and k is the fuel type. The activity level referred to the number of buildings and the energy intensities of each activity referred to the energy use per building. These variables are presented in Ugwoke, Becchio et al. (2020) and Ugwoke, Sulemanu et al. (2020). The demand subsectors encompassed four branches, which were further disaggregated into three energy end use branches namely lighting, electric appliance and space cooling (see Fig. 5). Based on the local characteristics of the rural built environments, the disaggregation, encompassed residential and non-residential demand sectors of which the residential sector accounted for more than 85 % of the total energy demand across the two communities (Ugwoke, Sulemanu et al., 2020). The non-residential sectors at less than 15 % of the total energy across the two communities accounts for the productive energy use sector and includes the community (health centre and school), commercial (tailor shop, provision store, viewing centre, beauty shop and barber shop) and agriculture sectors (grinding and milling shop).

The scenario analysis adopted four scenarios namely the reference

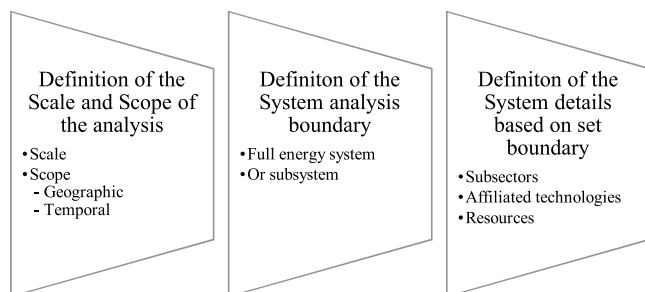


Fig. 2. Steps followed for the development of the LEAP model adapted from (Rogner, 2017).

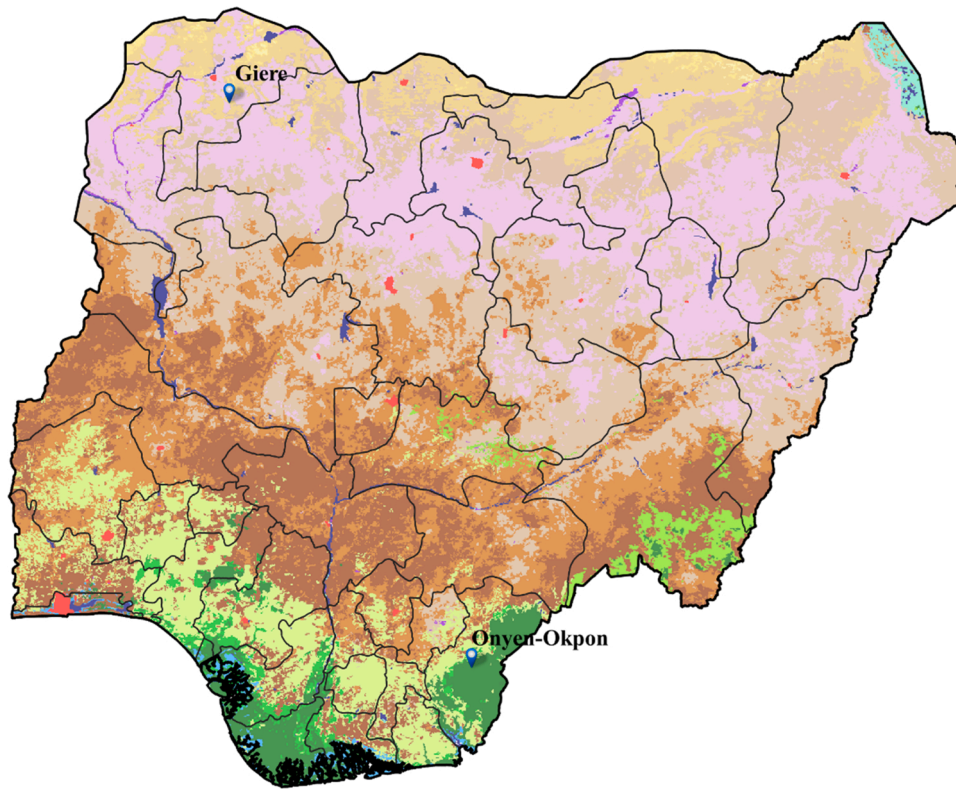


Fig. 3. Map of Nigeria depicting the two communities (source: from the authors).

(REF) scenario and three DSM scenarios. Based on the study carried out for estimating community energy demand and demand reduction potential (Ugwoke, Sulemanu et al., 2020), the annual building energy use provided the endogenous demand in the base year (2015) and the composite scenarios namely DSM 1 based on advanced retrofit-IEQ_I, DSM 2 based on advanced retrofit-IEQ_{II} and DSM 3 based on advanced retrofit-IEQ_{III}. The REF scenario inherited the base year attributes, while the DSM scenarios adopted the composite scenarios for demand reduction targets in the end year (2040). These scenarios are presented in Table 1. The demand was projected using the different sectoral annual growth rates in NEMP which include 3.16 % for the residential sector, 12 % for both commercial and community sectors and 6 % for the agriculture sector (Energy Commission of Nigeria (ECN) (2014)).

3.2.2. The energy supply (transformation) analysis

The transformation processes constituted the energy supply system and optimizing electricity generation options. The transformation module (see Fig. 5) based on Nigeria's REA mandate, prescribed the deployment of mini-grid energy systems (Yakubu, Ayandele, Sherwood, Olu, & Graber, 2018). The systems are RE based and considered the availability of indigenous RE resources. Information on the different system parameters pertaining to the transformation modules were obtained from diverse sources and presented in Tables 2 and 3, while the fuel costs are presented in Tables 4 and 5. A detailed description of the transformation system is available in Heaps (2020) and Ouedraogo (2017). The net energy consumption (ET) for transformation is calculated as follows (Emodi et al., 2017);

$$ET_l = \sum_m \sum_n ETP_{m,n} * \left(\frac{1}{f_{m,n,l}} - 1 \right) \quad (2)$$

where ETP is the product from the transformation process (electricity based on the power plants), f is the energy transformation efficiency, l is the type of primary energy, n is the technology, and m is the type of secondary energy.

For each process p :

$$Input_p = \frac{Output_p}{Efficiency_p} \quad (3)$$

For the distribution module:

$$Efficiency_p = 1 - Losses_p \quad (4)$$

where Input is the fuel or feedstock used in the process of power generation, Output is the amount of electricity generated and Efficiency represents the efficiency of the power plant.

The GHG emissions from energy transformation (GHG_T) are calculated as follows;

$$GHG_T = \sum_l \sum_m \sum_n ETP_{m,n} * \frac{1}{f_{m,n,l}} * EF_{m,n,l} \quad (5)$$

EF is the emission factor from one unit of primary fuel type, l used to produce secondary fuel type m through equipment n . ETP and f are as specified above.

There was no information on historical production and exogenous capacity in these communities. The technological electricity generation options are presented in Table 3. The possible mini-grid system configurations included the stand-alone system technologies (solar photovoltaic (PV) plant, biomass gasification (biomass combined heat and power (B-CHP)) plant and diesel genset) and the optimization (combined hybrid) option (solar PV + biomass plant + diesel system) based on LEAP's optimization capabilities. Hybridizing and co-optimization of these technologies shows particular promise (Jones & Leibowicz, 2021) and this has already been demonstrated (Jamshidi & Askarzadeh, 2019; Movahediyani & Askarzadeh, 2018). These represented the most promising technological options based on the viable RE resource availability and therefore the wide technology dataset available in LEAP was limited to these primary resources and related technologies.

The system load curve was calculated taking into consideration the hourly demand profile and created time slices within a year. The systems

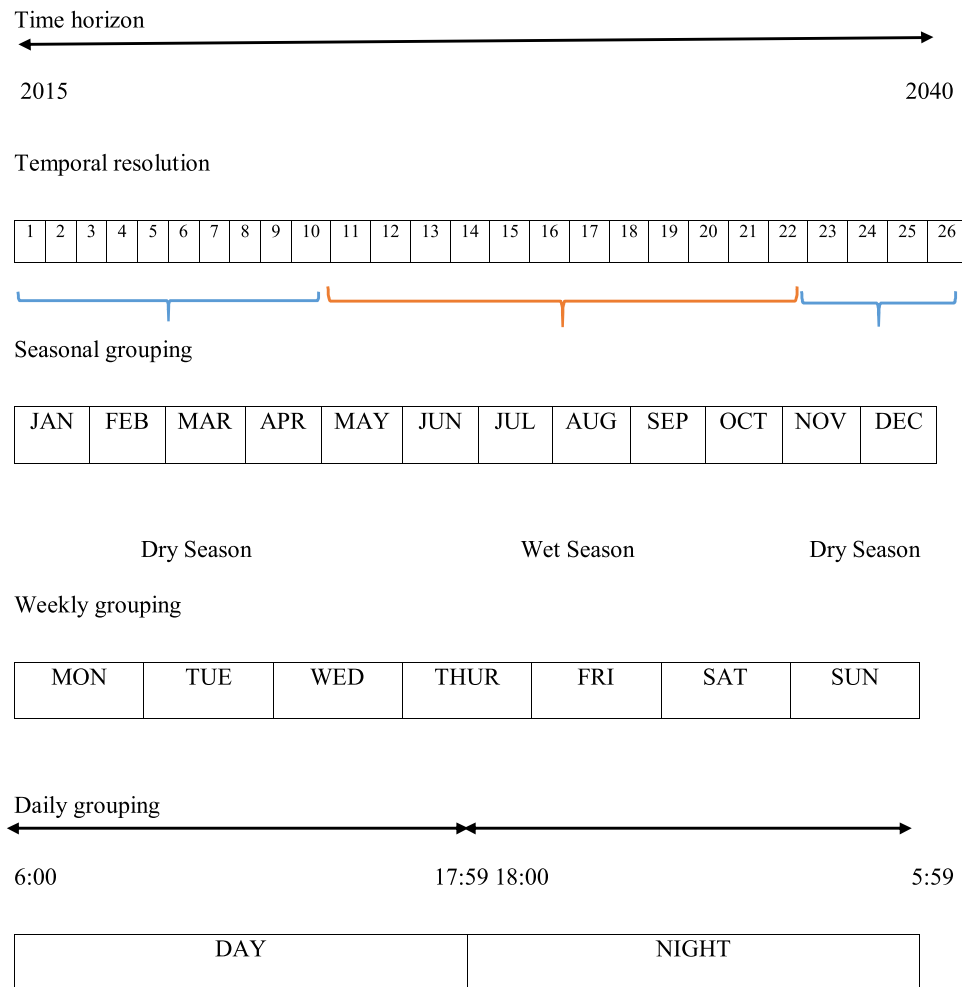


Fig. 4. Temporal scope for the analysis.

were on the order of 2–5 MW,¹ therefore it was assumed that size variation within this limit would not affect significantly capital cost and efficiency of technologies. Supply-side EE measures are incorporated into the modelling framework through reduction of the distribution losses in the DSM scenarios. The TED² platform provided technology-specific emissions data for the estimation of the global warming potential of the scenarios. The optimization analysis was employed to determine the optimal supply system configuration and the associated system variables including the required capacities (the type and schedule of capacity addition and retirement), dispatch order, and resource requirements.

The demand cost is the cost per activity used to specify non-fuel cost per unit of activity (Heaps, 2020). The distribution losses entail the upstream electricity losses encountered in LEAP's distribution module used in distribution to the end-users. Planning reserve margin is used to designate endogenous capacity addition in LEAP. It is the amount of firm generation capacity minus the maximum annual demand expressed as a ratio of maximum annual (Heaps, 2020).

The process efficiency describes the percentage ratio of energy

outputs to feedstock energy inputs. The lifetime is the economic life of a generation technology or electric plant. The capacity credit describes the portion of the rated capacity of a power plant that can be considered firm.³ The maximum availability (or maximum capacity factor) of a generation technology or electric plant is the ratio of the maximum energy produced to what would have been produced if the plant or generator produced at full capacity for a given period, usually it is expressed on yearly basis (Heaps, 2020).

Capital cost is the total (i.e., non-annualized) capital cost per unit of capacity of each generation technology or electric plant such that considers the equipment costs (including engineering), civil, construction and physical contingency costs (ESMAP, 2007b). No effect of size is considered in the capital cost since technology size for the specific applicant is almost constant (2–5 MW). For the PV system and considering fully autonomous operation for uninterrupted (100 %) off-grid applications, this comprised the solar PV module cost and balance of system (BoS) hardware⁴ and non-hardware⁵ cost (IRENA, 2016). Capital

¹ This expatiates the size of endogenous capacities for the generation technologies.

² TED reference reports from different institutions including the Intergovernmental Panel on Climate Change (IPCC), the U.S. Department of Energy, and the International Energy Agency. TED also provides qualitative information on wide range of energy technologies including the cost-effectiveness and key environmental issues for a wide range of energy technologies (Heaps, 2020).

³ "Firm" implies the plant capacity that can be readily available and accessible.

⁴ The BoS hardware costs include the costs of battery, inverter, racking, wiring and cables, control, monitoring and measuring hardware, duty and transportation.

⁵ The BoS non-hardware costs encompass soft costs such as system design and procurement costs, subsidies (applications, fees, etc.) permitting applications (with utility provider and other authorities), financing and contract etc. (IRENA, 2016).

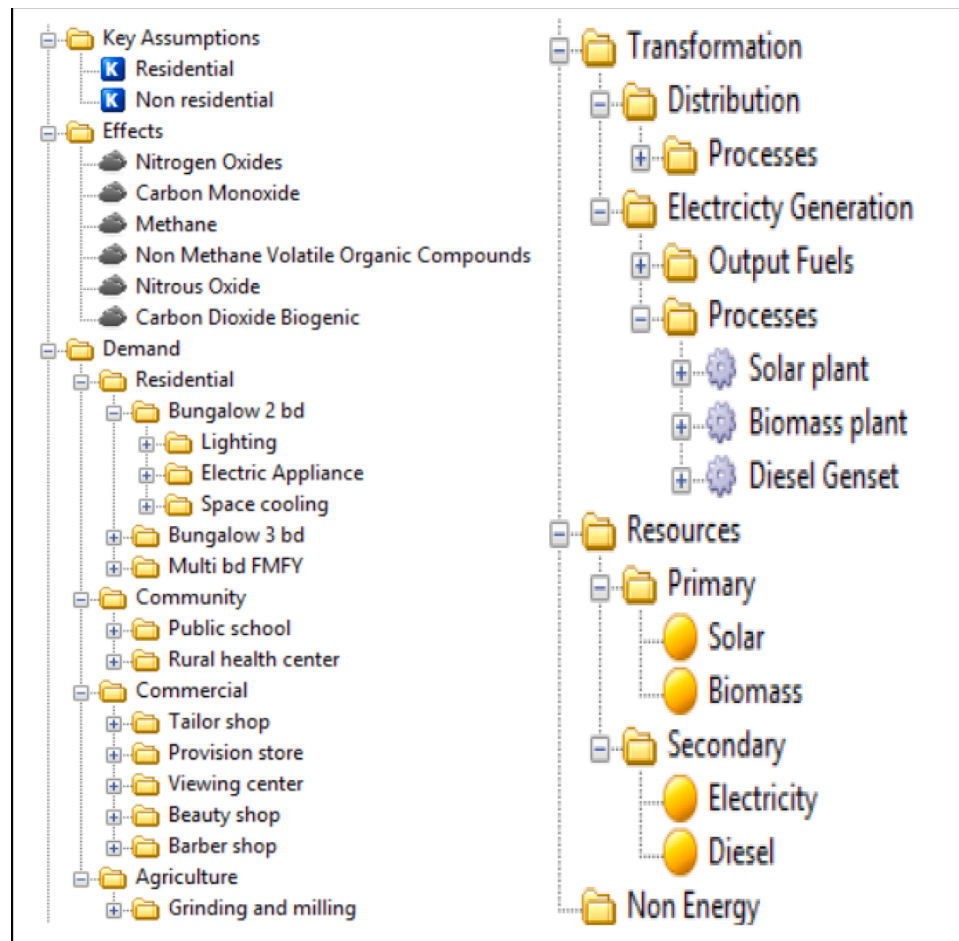


Fig. 5. Data structure for community-scale LEAP model (Note: bd-bedroom, FMFY-face me I face you).

Table 1

Demand scenarios for the two communities.

	Onye-okpon				Giere			
	REF 2015	DSM 1 2040 Targets	DSM 2	DSM 3	REF 2015	DSM 1 2040 Targets	DSM 2	DSM 3
Residential sector demand (GWh)	4.452	3.844	2.689	1.953	1.226	0.718	0.583	0.490
Commercial sector demand (GWh)	0.063	0.048	0.041	0.037	0.009	0.006	0.006	0.005
Community sector demand (GWh)	0.169	0.131	0.127	0.118	0.173	0.131	0.123	0.115
Agriculture sector demand (GWh)	0.240	0.128	0.128	0.128	0.024	0.013	0.013	0.013

Source: (Ugwoke, Becchio et al., 2020; Ugwoke, Sulemanu et al., 2020).

Table 2

System details.

System Details	2015	2040
Distribution losses (%)	25 ^a	20 ^a
Planning Reserve Margin (%)	10 ^{a,b}	
Interest rate (%)	5 ^c	
Demand Costs (\$/kWh)	0.6 ^{d,e}	0.2 ^{d,e}

^a IRENA (2018a).

^b Ouedraogo (2017).

^c (The World Bank, 2018) based on World Bank Group technical guidance note. The same rate was applied throughout the entire duration of the planning period. Sensitivity analysis was carried out given the disparities between the values of the discount rate (see Section 4.5).

^d Rural Electrification Agency (REA) (2017a).

^e Tudy and Romising (2017).

costs reduction rate described the reductions of the capital costs of generation technologies based on technology forecasts via learning curves and technological learning (IRENA, 2018b; Roche, Ude, & Ofoegbu, 2017).

Fixed operating and maintenance (O&M) costs are those incurred regardless of the energy produced by a generation technology or electric plant (e.g. asset and site management, overhead costs, insurance, security, equipment costs, labour, materials, contract services for routine O&M, and administrative and general costs). The variable O&M costs are those incurred per unit of energy produced (e.g. costs of consumables (excluding fuel costs), preventative and corrective maintenance costs, short run incremental costs (for fuel and labour) (California ISO, 2019; Heaps, 2020).

From Tables 4 and 5, the biomass is a much less expensive fuel than diesel. There are several restrictions to biomass adoption including access to fuel in some areas, lack of local technical capacity for project design, construction, operation and maintenance. Although these

Table 3
Generation technologies parameters (base year 2015).

Generation Technologies Details	Solar and Battery	Biomass Gasification	Diesel Generator
Process Efficiency (%) ^a	100 ^{a,b}	30 ^{a,b}	30 ^{a,b}
Lifetime (Years)	25 ^a	30 ^a	20 ^a
Capacity Credit (%)	20 ^{b,c,d}	40 ^{b,c,d}	40 ^{b,c,d}
Maximum Availability (%)	20 ^{b,c,d}	80 ^{b,c,d}	80 ^{b,c,d}
Capital Costs (\$/kW) ^e	6000 ^a	2500 ^a	600 ^a
Capital costs Reduction Rate (%)	−5.9 ^{e,f}	−5.8 ^{e,g}	–
Fixed O&M Costs (\$/kW)	30 ^{a,f}	82 ^{a,f}	21 ^{a,f}
Variable O&M Costs (\$/MWh)	20 ^a	20 ^{a,f}	55 ^{a,f}

^a Energy Sector Management Assistance Program (ESMAP, 2007a).

^b International Renewable Energy Agency (IRENA, 2018a).

^c Renewable Energy Policy Network for the 21 st Century (REN 21, 2019).

^d Intergovernmental Panel on Climate Change (IPCC, 2011).

^e International Renewable Energy Agency (IRENA, 2018b).

^f Nigerian Economic Summit Group (NESP) (Roche et al., 2017).

^g International Renewable Energy Agency (IRENA, 2012). It should be noted that this is based on IRENAs projections and this is conservative compared to those provided by some countries e.g. China in 10.96 % (Liu & Li, 2016).

^{*} Capital cost and efficiency are kept constant because the size range of technology is narrow (2–5 MW) and therefore size effect on those variables has been considered negligible.

Table 4
Diesel fuel costs.

Diesel Costs (\$/kWh)	2015	2020	2040
Lower bound	0.040 ^a	0.062 ^{a,b}	0.087 ^a
Higher bound	0.046 ^a	0.072 ^{a,b}	0.101 ^a

^a (IRENA, 2018a): based on high-level benchmarks for fossil fuels from the International Energy Agency (IEA) New Policies Scenario in the 2016 World Energy Outlook and corresponding prices ratios from the West African Power Pool (WAPP) Master Plan.

^b (Nigeria National Bureau of Statistics (NBS) (2019)): the projected costs for 2020 correlated with the current Nigerian diesel costs.

Table 5
Biomass fuel costs.

Biomass Costs (\$/kWh)	2015
Lower bound	0.0036 ^a
Higher bound	0.0180 ^a

^a Constant biomass costs were assumed for the planning period.

Source: (IRENA, 2018a; Roche et al., 2017).

biomass technologies are available, the practice involves retrofitting diesel internal combustion engine (ICE) plants into biomass integrated gasification plants, which involve both more capital and far more technical competence than conventional diesel generators.

3.2.3. Integrated cost benefit analysis

This analysis avails a comparative overview of the costs and benefits of the composite scenarios compared to the reference scenario (Heaps, 2020). The costs of the different scenarios were analysed and combined with cost-benefit comparisons by considering various cost components (demand costs, transformation costs (capital and O&M costs), resources (primary resource and secondary fuel costs) and environmental or externality costs (the social costs of different GHGs (SC-GHGs)), (Emodi et al., 2017).

The SC-GHGs are metrics adopted for estimating the benefits associated with incremental reductions of GHGs emissions (Marten & Newbold, 2012). They are used to estimate the monetized damages

associated with the emissions of GHGs (Interagency Working Group on Social Cost of Greenhouse Gases, 2016) using the values previously developed in the literature and summarized in Table 6. It should be noted that these social costs are all conservative as they do not include the potential for large scale climate-related liability over the long term (Allen, 2003; Heidari & Pearce, 2016).

4. Results and discussion

This section provides the results obtained from the analyses performed. This consists of six parts: results obtained from the demand projections, optimization analysis, supply projections, GHG emission projections, integrated cost benefit analysis and the policy implications for the two communities.

4.1. Demand projections

The demand projections for the scenarios and end uses are presented in Figs. 6 and 7. In 2015, the community energy demand was 4.92 GW h and 1.44 GW h for Onye-okpon and Giere, respectively. All scenarios exhibit a continuous increase, but with different growth rates (see Section 3.2.1). In the REF scenario, the demand is projected to be three and four times higher in 2040 than 2015 for Onye-okpon and Giere, respectively. This will amount to 14.64 GW h for Onye-okpon and 5.87 GW h for Giere in 2040.

The DSM scenarios provide demand reduction at varying degrees due to the progressive adoption of demand side management by including energy efficiency and advanced retrofit measures across the different end use areas (i.e. lighting, appliances and space cooling). IEQ requirements are also considered to provide adaptive comfort strategies for the local built environment. For Giere, by 2040, the demand will be 3.95 GW h, 3.51 GW h and 3.17 GW h for the DSM 1, DSM 2 and DSM 3 scenarios, respectively. In Onye-okpon, by 2040, the demand will be 11.97 GW h, 9.25 GW h and 7.44 GW h for the DSM 1, DSM 2 and DSM 3 scenarios, respectively. The overall demand reduction across the two communities will be in the range of 18 %–49 % lower than the REF scenario. The bulk of the energy demand is attributed to space cooling and this is contingent on the IEQ (Chan & Chau, 2021). Lighting presents the largest potential for relative demand reduction (81 %–85 %), followed by appliances (48 %) and space cooling (4 %–45 %). There is a continuous decrease but at different rates due to the different indoor microclimate requirements (ASHRAE 55:2010, ISO 7730 and CEN EN 15251 (2007). IEQ control avails demand reduction while attaining levels of thermal comfort and indoor air quality within reasonable and acceptable standards for wellbeing and overall productivity. These reductions result in energy savings, economic remunerations and other benefits (see Section 4.5). The DSM 3 scenario provides the largest demand reduction due to its less stringent requirements for IEQ.

From the sectorial and sub-sectorial demand projections for the REF scenario in Figs. 8 and 9 and Table 7, the residential sector accounted for

Table 6
Social costs of GHGs (SC-GHGs).

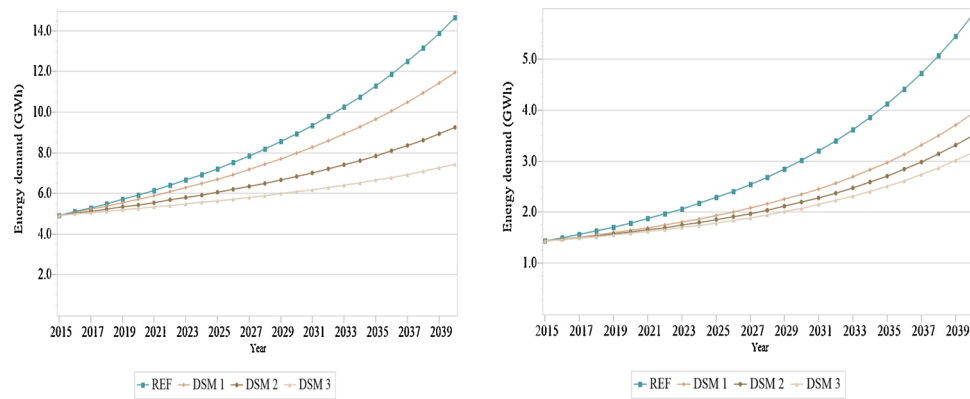
GHGs	Social costs (\$/tonne)
Carbon dioxide (CO ₂)	220 ^a
Nitrous oxide (N ₂ O)	13,000 ^{b,c}
Methane (CH ₄)	1000 ^{b,c}
Non-Methane Volatile Organic Compounds (NMVOC)	339 ^d
Nitrogen oxides (NO _x)	88 ^d
Carbon monoxide (CO)	946 ^d
Sulphur dioxide (SO ₂)	713 ^d

^a Based on integrated assessment model (IAM) that considers economic growth rate and economic disparity among countries (Moore & Diaz, 2015).

^b Marten and Newbold (2012).

^c (Interagency Working Group on Social Cost of Greenhouse Gases (2016).

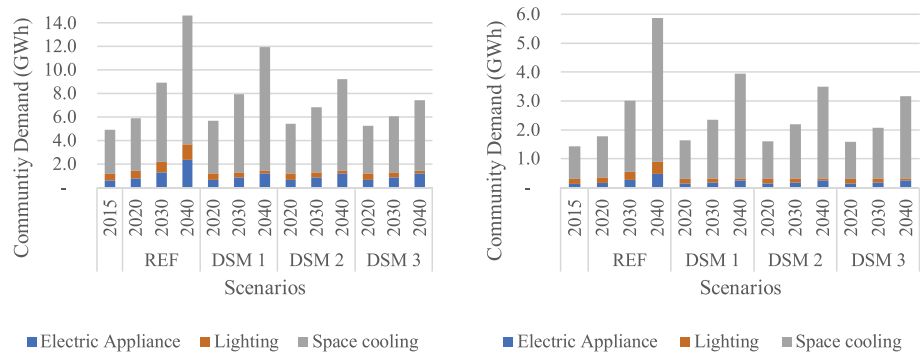
^d Howells et al. (2005).



A) Onye-okpon

B) Giere

Fig. 6. Energy demand projection by scenarios.



A) Onye-okpon

B) Giere

Fig. 7. Energy demand-end use projection.

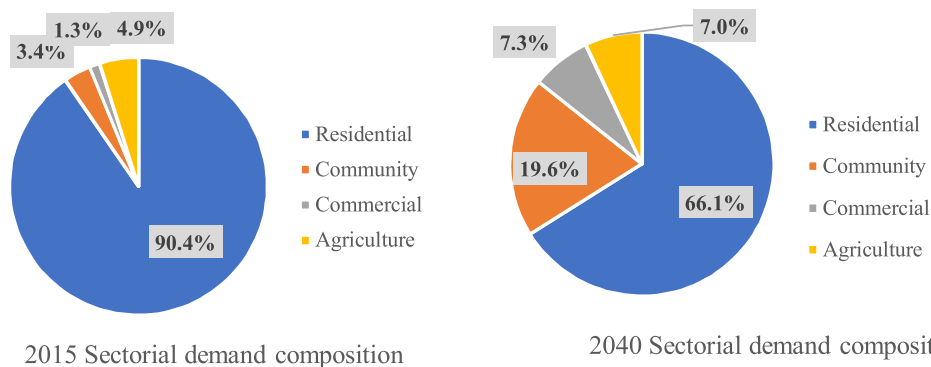


Fig. 8. Sectorial energy demand projection-REF scenario Onye-okpon REF scenario.

more than 85 % of the total demand across the two communities in 2015. The remaining demand was split across the remaining productive energy use sectors in varying proportions. At 5 % of the total demand, the agriculture sector emerged the second largest demand sector in Onye-okpon while the community service sector was the second largest demand sector in Giere at 12 %. The composition of the sub-sectorial demand also varied accordingly owing to the different distribution of their built environments. The sectorial demand composition remained largely unchanged for Onye-okpon in 2040 although the share of the

residential demand declined by 24 % from 90 % in 2015. In Giere, this mix is projected to change by 2040 with the community service sector demand (50 %) overtaking the residential sector demand (46 %). These composition variations could materialise into added economic value and overall wellbeing based on increased energy consumption by non-residential and productive sectors especially income generating sectors (agriculture and community) (Rocco et al., 2020).

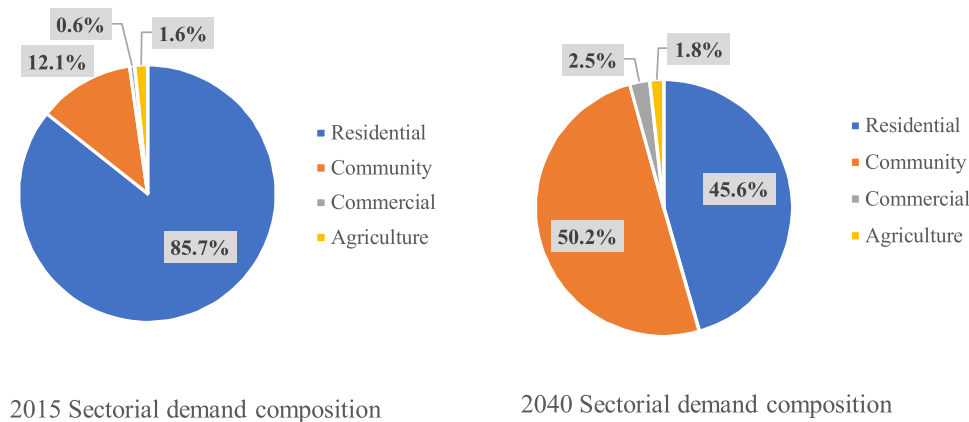


Fig. 9. Sub sectorial energy demand projection: Giere REF scenario.

Table 7

Composition of the projected sub sectorial energy demand: REF scenario.

Sub sectorial demand composition (%)	Onye-okpon		Giere	
	2015	2040	2015	2040
Bungalow 2 bedroom	24.89	18.21	41.36	22.01
Bungalow 3 bedroom	9.16	6.70	19.39	10.32
Multi bedroom FMFY	56.38	41.23	24.90	13.25
Barber shop	0.06	0.34	0.10	0.40
Beauty shop	0.06	0.34	0.07	0.31
Provision store	1.00	5.71	0.33	1.36
Public school	2.70	15.44	9.12	37.92
Rural health centre	0.72	4.12	2.95	12.28
Tailor shop	0.12	0.66	0.09	0.39
Viewing centre	0.04	0.24	0.00	0.00
Grinding and milling	4.87	7.02	1.68	1.76

4.2. Optimization analysis

For the optimization analysis, the cumulative discounted social costs⁶ and the global warming potential (GHGs emissions) for the different configurations of generation technologies are presented in Figs. 10 and 11. From the total social cost structure, the bulk of the cost was attributed to fuel import. The solar only option provided the cleanest and cheapest supply technology but is limited to only one resource and subject to intermittent supply. The biomass only option was the second most expensive option but limited to only one resource unlike the optimization option which embodies a mix of the different generation technologies in varying proportions. The optimization option provides an optimal supply system configuration and informs the required capacity, and generation structure. This optimal option is then adopted and applied to the elicited scenarios to project the energy supply and GHGs emission for the entire planning horizon.

Sensitivity analysis were performed on the fuel cost bounds for the diesel and biomass plants. The results are provided in Figs. 12 and 13, which depict variations in the cumulative social costs and GHGs emissions across both communities. The GHGs emissions slightly increased with the lower bound fuel costs than the higher bound fuel costs. This implied that with the optimization option,⁷ there is the tendency to increase the diesel plant utilization when the fuel costs are lower. The diesel-only option was the most expensive and produced the highest GHG emissions and social costs. This persisted for both bounds of fuel

costs with some variations in the total costs of some technologies.

4.3. Supply projections

The projected supply for all scenarios is shown in Fig. 14 and the generation structure in Fig. 15. In 2016, with overall generation capacity of 5 MW and 1.5 MW, the supply arrived at 6.56 GW h and 1.91 GW h for Onye-okpon and Giere, respectively. The share of the generation structure is 40 %–44 % solar, 38 %–49 % biomass and 11 %–18 % diesel. In the REF scenario, the supply is projected to be three and four times higher in 2040 than 2015 for Onye-okpon and Giere, respectively. This will amount to 19.54 GW h for Onye-okpon and 7.83 GW h for Giere in 2040 and 100 % renewable based largely powered by solar (99 %) which will develop at a relatively fast rate. This may be attributed to technological constraints and resource development potential for biomass (Chen et al., 2020). The diesel plant will be phased out before 2030 and 2040 for Onye-okpon and Giere, respectively. The overall generation capacity increased to 18.9 MW for Onye-okpon and 7.6 MW for Giere.

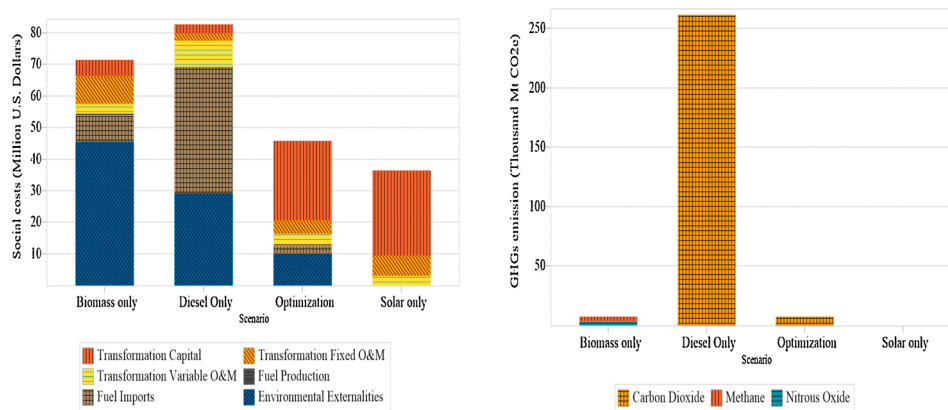
The generation structure of the DSM scenarios is mirrors that of REF. For Giere, by 2040, the supply will be 4.93 GW h, 4.38 GW h and 3.96 GW h for DSM 1, DSM 2 and DSM 3 scenarios, respectively. In Onye-okpon, by 2040, the supply will be 14.96 GW h, 11.57 GW h and 9.30 GW h for DSM 1, DSM 2 and DSM 3 scenarios, respectively. The demand reduction trend translates to the supply such that by 2040 the supply reductions are in the range of 23 %–52 % lower than the REF scenario. This is because supply side efficiency measures such as reduction of distribution losses are also incorporated with the elicited demand side efficiency measures. Again, the DSM 3 scenario will provide the largest supply reduction. There are commensurate capacity reductions across the communities. For Giere, by 2040, the overall generation capacity will be 4.8 MW, 4.3 MW and 3.9 MW for the DSM 1, DSM 2 and DSM 3 scenarios respectively. In Onye-okpon, by 2040, the overall generation capacity will be 14.4 MW, 11.1 MW and 9.1 MW for the DSM 1, DSM 2 and DSM 3 scenarios respectively. The overall capacity reduction follows the supply reduction trend.

4.4. Greenhouse gases (GHGs) emissions

The GHGs emissions and the GHG mix for all scenarios are shown in Figs. 16 and 17. The trends in the energy supply system translates into GHG emissions. The GHGs emissions across the two communities will increase irrespective of the scenario. This is mostly due to the increase in power generation associated with the increase of energy demand. Beyond 2030, the cumulative emissions become plateaued due to the increased deployment of solar PV systems in the energy supply mix. All through, CO₂ makes up the largest proportion in the GHGs emission mix, contributing more than 80 % of emissions. In the REF scenario, by 2040, the GHG emissions will reach 7309 and 2157 metric tonnes of CO₂

⁶ The social costs represents the overall costs to society as opposed to the particular costs of production seen by producers or consumers (Heaps, 2020).

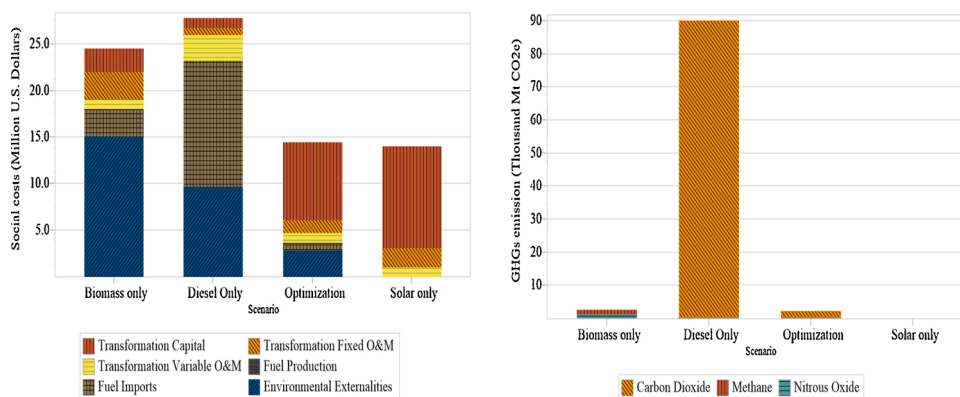
⁷ Based on LEAP's optimization capabilities such that LEAP automatically decides what combination of power plants will meet demands at the lowest cost.



A) Cumulative discounted social costs

B) Cumulative GHGs emission

Fig. 10. Cumulative Discounted social costs and GHGs emission in 2040: Onye-okpon.



A) Cumulative discounted social costs

B) Cumulative GHGs emission

Fig. 11. Cumulative discounted social costs and GHGs emission in 2040: Giere.

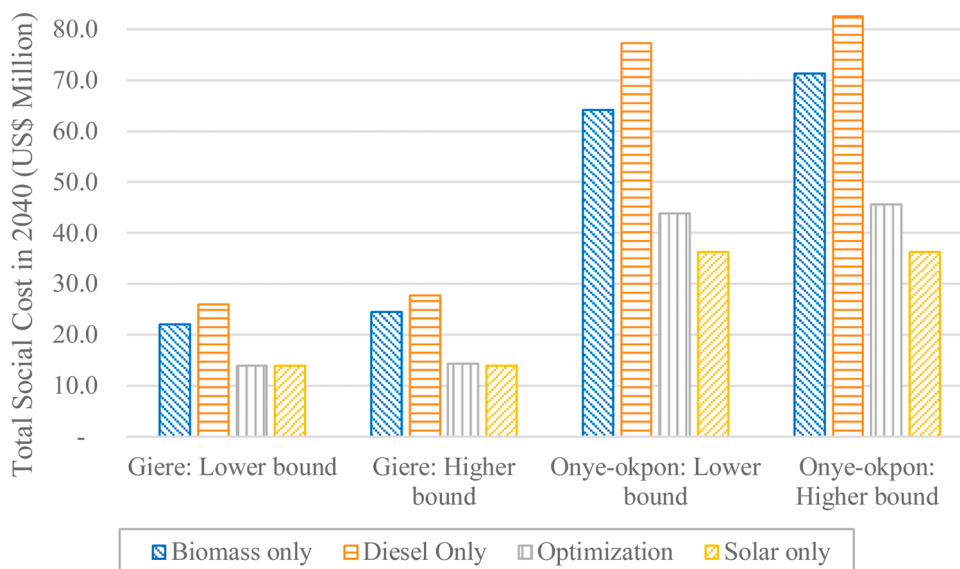


Fig. 12. Sensitivity analysis on fuel costs bound: Cumulative discounted social costs.

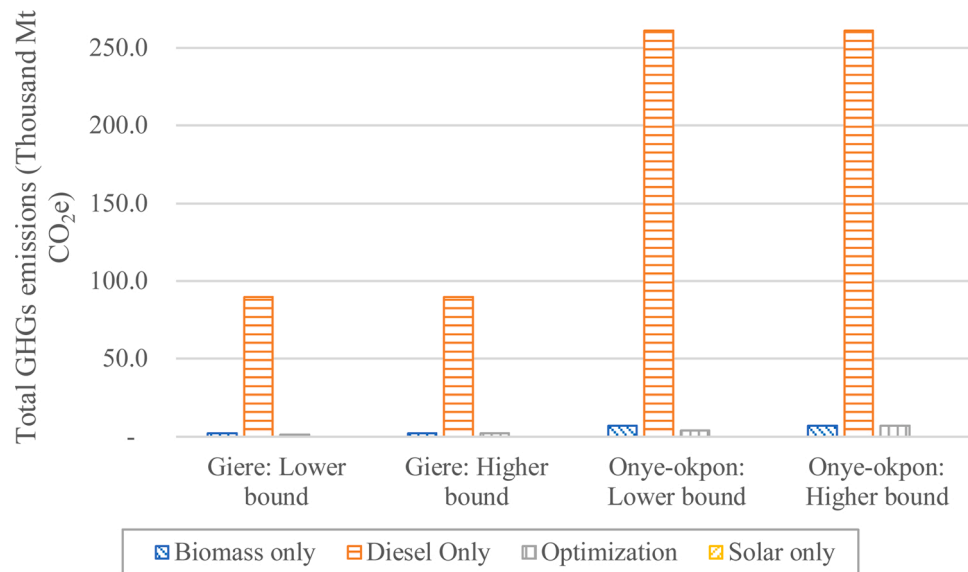


Fig. 13. Sensitivity analysis on fuel costs bound: Cumulative GHGs emission.

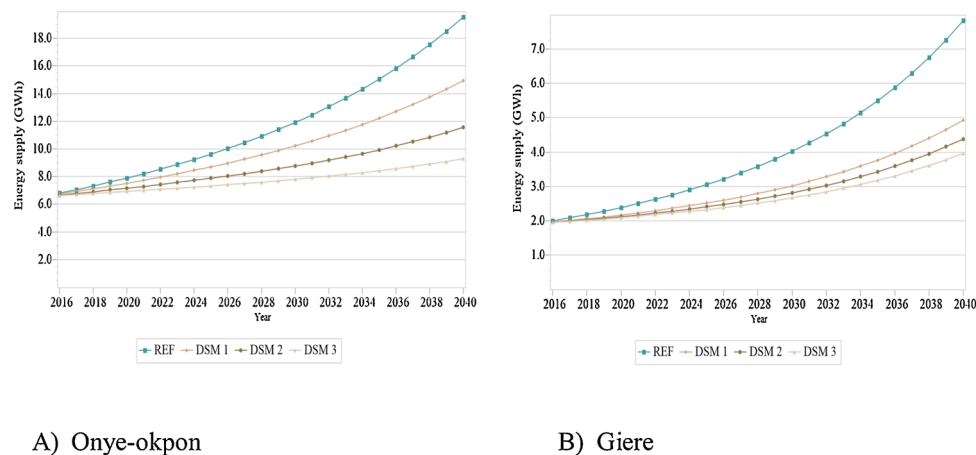


Fig. 14. Energy supply projection by scenarios.

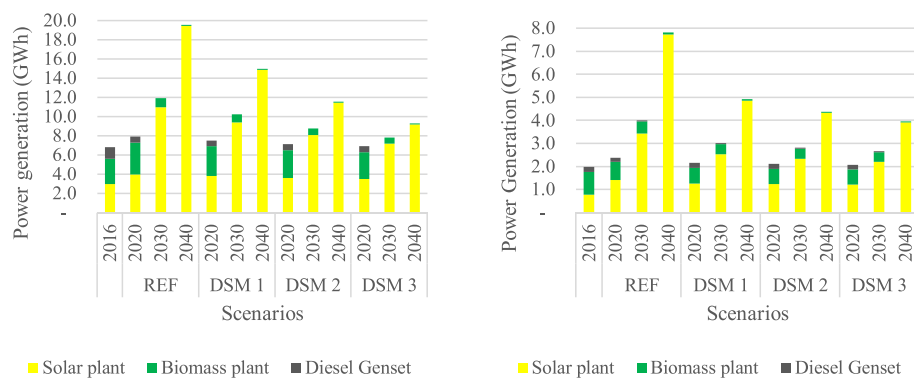


Fig. 15. Power generation: technology mix projection.

equivalent (Mt CO₂e) for Onye-okpon and Giere, respectively. The composite scenarios will avail emission reductions in the range of 0.3 %–16 % lower than the REF scenario by 2040. The DSM 3 scenario will

provide the largest GHG reduction with emissions at 6,111 and 2,083 Mt CO₂e by 2040 for Onye-okpon and Giere, respectively.

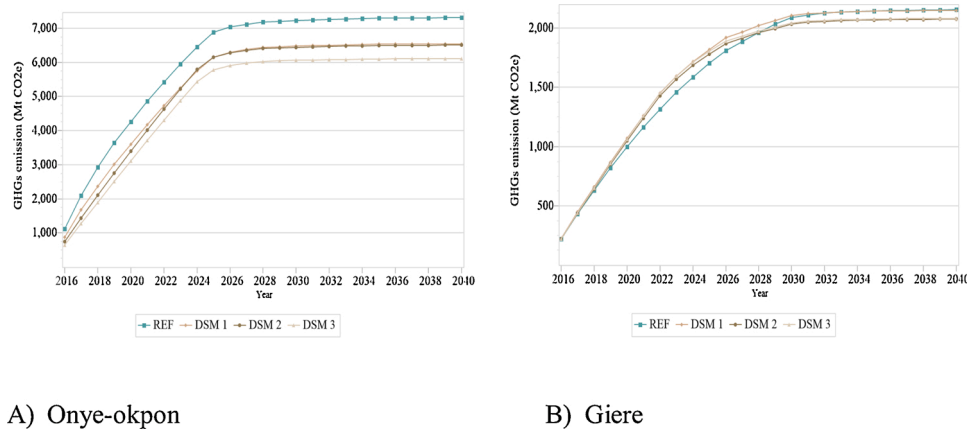


Fig. 16. Cumulative GHGs emission by scenarios.

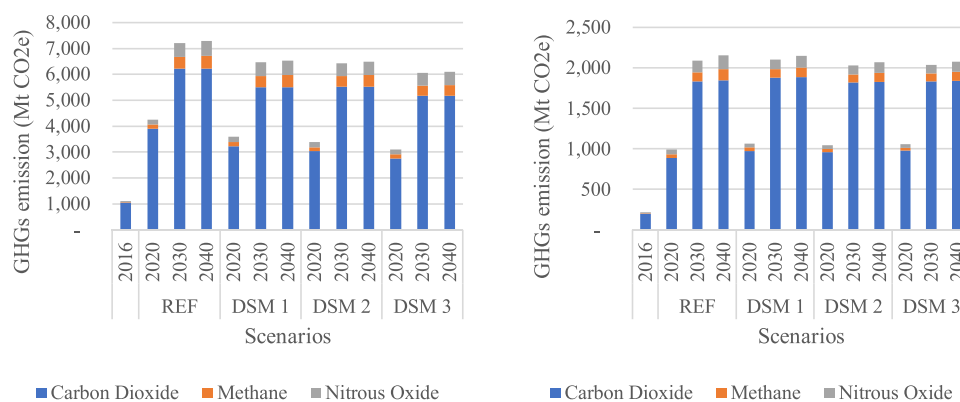


Fig. 17. Cumulative GHGs emission mix projection.

4.5. Integrated cost benefit analysis

The results of the integrated cost-benefit analysis are presented in Figs. 18 and 19. At a discount rate equivalent to the interest rate, these results showed cost ramifications of the composite scenarios relative to the REF scenario (Heaps, 2020) and the gains accrued from avoiding GHGs emission. The overall costs of these scenarios are lower than those of the REF scenario all through the different segments from demand to resource. This imply benefits⁸ at varying degrees across the scenarios.

The generation (transformation) and residential demand sects provided the largest benefits. Based on the NPV, the DSM 3 scenario accrued the highest benefits of US \$16.3 million and US \$5.4 million for Onye-okpon and Giere, respectively. However, the GHGs savings was more in the DSM 2 scenario than in the DSM 3 scenario for Giere. In Giere, the benefit of avoiding GHGs emissions arrived at US \$629,614/Mt CO₂e, US \$59,434/Mt CO₂e and US \$72,793/Mt CO₂e for DSM1, DSM2 and DSM 3 scenarios respectively. In Onye-okpon, the benefit of avoiding GHGs emissions arrived at US \$8,268/Mt CO₂e, US \$15,381/Mt CO₂e and US \$13,609/Mt CO₂e for DSM1, DSM2 and DSM 3 scenarios respectively.

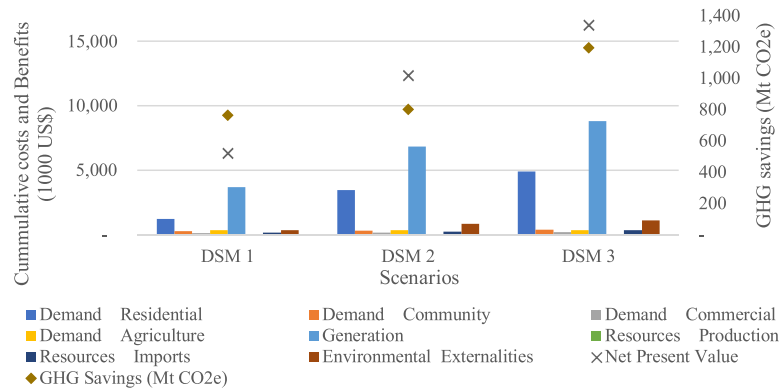
⁸ “Benefits” are accrued when the cumulative costs of the scenario are lower than those of the reference scenario, otherwise “Costs” are incurred.

The results from the sensitivity analysis performed on the NPV benefits for different discount rates⁹ and fuel costs bounds are shown in Fig. 19. There are drastic reductions in the benefits accrued across the scenarios with increased discount rates. This trend persists for the bounds of the fuel costs.

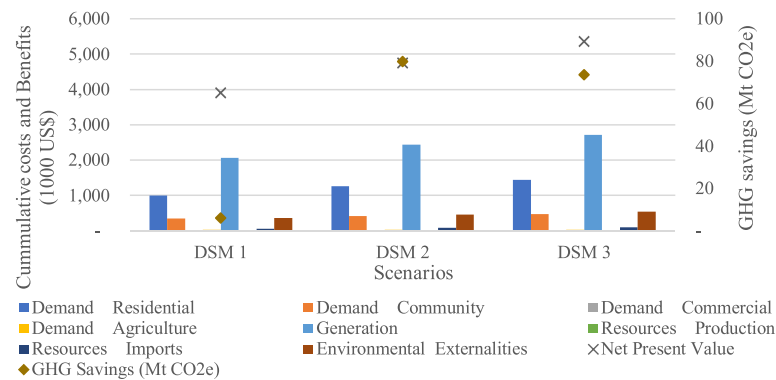
4.6. Future energy system

The representation of the future energy systems for the two communities are presented in Figs. 20 and 21. These summarise the energy demand, conversion and production across the communities to depict the energy flows through different elements of the energy system and demonstrates the likely behaviours of these system elements. By 2040, the energy production would be 100 % RE based and mostly solar. The energy conversion is encumbered by huge distribution losses that eclipse the energy demand of most of the demand sub-sectors. There are different trends across the communities for high-demand sectors and these could inform the landscape for designing local energy policies and defining strategies for prioritising energy access initiatives.

⁹ This was done to capture extreme cases and avail wide margins to showcase the extremities at both ends of the spectrum (very low and very high cases) and ultimately future proof the obtained results (Old School Value, 2020).



A) Onye-okpon



B) Giere

Fig. 18. Cost-benefit results: strategy's impact analysis.

4.7. Policy implications

The level of synergies attainable by integrating both CP and EP strategies and the obtained results are interesting from an energy policy localisation perspective since there is the possibility to define and reach multiple targets, goals and impacts (Chen & Chen, 2019). Therefore, this strategy integration could provide an effective policy approach to better position communities to harness their inherent potential towards mitigating climate change (Zhou & Chen, 2019).

Given the magnitude of the rural population in the developing world and the opportunities for deploying decentralized energy systems to assuage their energy access problems, insights from the C-SEP would foster the design of effective policy initiatives. Therefore, the policy implications for the obtained results are expatiated below together with some energy policy issues being addressed;

- In the energy consumption structure, the bulk of the consumption is tied to the residential sector. The structure becomes somewhat diversified in later years and this diversification is location dependent. To realise cleaner and sustainable socio-economic development brought on by the uptake of RE-based generation systems, energy

consumption for productive uses would need to increase (Edomah, 2019). Therefore, localised policies need to be designed to incentivize sectoral integration such that can integrate agriculture, community, and commercial sectors to the residential sector in order to offset overall energy expenditure for the community. These new policies and measures could be designed such that they allude to social inclusion and cultural appropriateness in order to improve livelihood security, increase resilience and reduce emissions. Insights from sectoral integration of the different demand sectors can then be translated into deriving regional integration policies that would strive to couple preferable local market. This could be by inter-connecting communities through combining their mutual RE resources and optimizing limited financial resources.

- C-SEP would foster the development of sustainable community energy master plans. These plans can be set-up without detracting from regional or national existing development plans. Location-specific plans would allow stakeholders to customize specific actions to drive community development pathways and understand the impacts of these actions within the bounds of different national and regional energy strategies. These plans would foster policy initiatives on capacity building and especially target largely marginalized

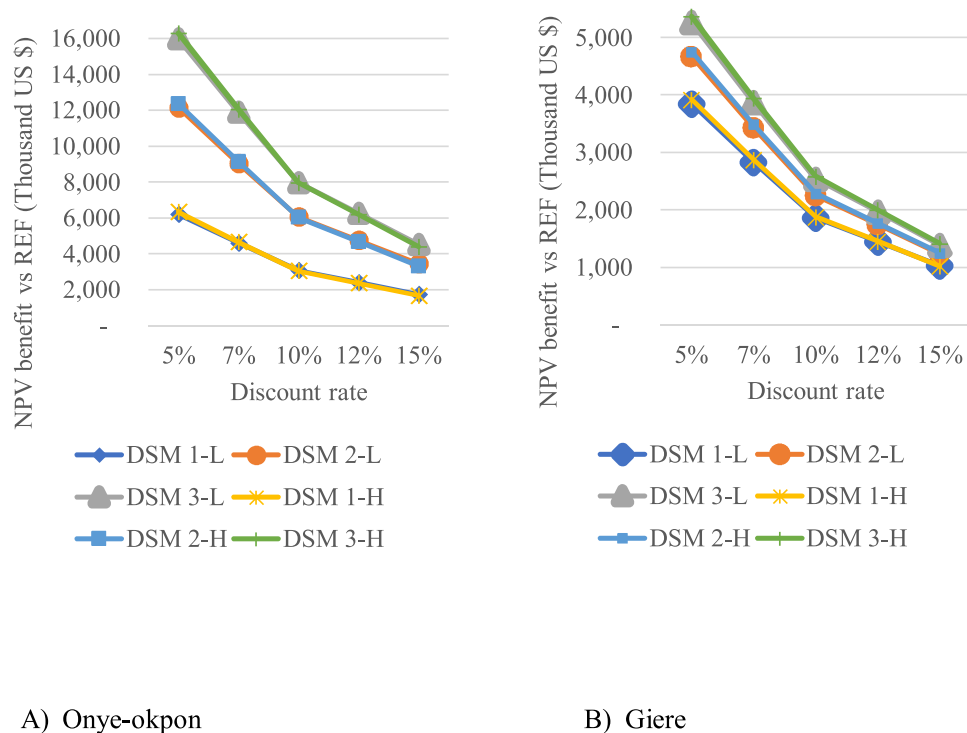


Fig. 19. Sensitivity analysis on discount rates and fuel costs bounds: NPV benefits.
Note: H-higher bound and L-lower bound of fuel costs.

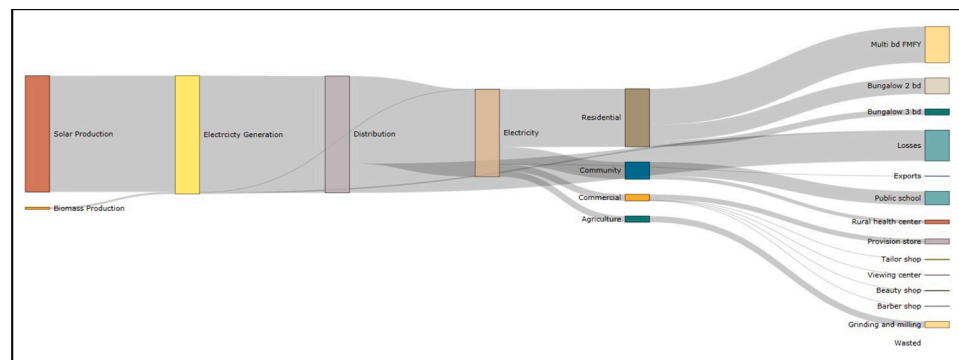


Fig. 20. Onye-okpon energy system: REF scenario in 2040.

genders and social groups by focusing on dedicated productive energy use sectors within their jurisdiction. These could include the women dominated establishments comprising tailor shop, provision store, and the beauty shops, to avail other empowerment opportunities. For social groups, localized policies could target schools to furnish child education initiatives and improve the ambience of the learning environment which translates to their growth, development and overall productivity.

- Considering climate change and the increased need for space cooling (Sustainable Energy for All, 2020), climate change adaptation needs to be considered with climate change mitigation strategies. This study coupled the IEQ criteria to buildings and equipment EE, rather than a more simplified approach in (Sharifi, Larbi, Omrany, & Boland, 2020). It also allowed for the consideration of the space cooling demand end use which is largely unexplored for rural electrification. However, it is an important aspect to be considered when developing policy for rural electrification. It is useful for setting demand reduction targets for adapting to different levels of expectation for indoor microclimate and defining the level of control

exerted thereof especially with the use of smart energy management systems. This would foster the resilience of the local built environment to climate change while decreasing the energy demand for space cooling. These results can inform the trajectory for implementing Nigeria's quota for the (Intended) nationally determined contributions ((I)NDCs) and commitment to peak CO₂ emissions around 2030 (Wienges, Bucher, & Hammer, 2016) from the local communities levels. In lieu of Nigeria's climate change policy response and strategy (NCCPRS) and the national adaptation strategy and plan of action for climate change Nigeria (NASPA-CCN), this study provides practical actions steps which can be translated to both state and national levels. As such, local policy makers could spur community participation and conveniently be at the forefront of driving local clean energy initiatives necessary to reach and/or surpass relative national policy targets in a cost optimal way. Also, by virtue of their proximity to the citizens, local authorities are better positioned to foster green education on environmental and sustainability benefits of clean energy technology development to raise social acceptance and awareness (Aithal & Rao, 2016). They can also

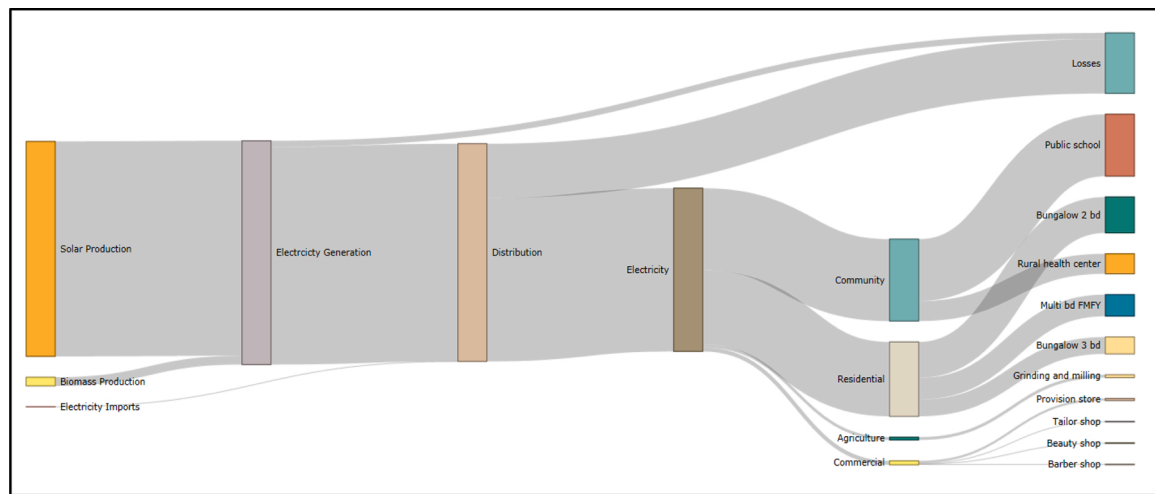


Fig. 21. Giere energy system: REF scenario in 2040.

facilitate collaborations among the citizens, local businesses and other relevant stakeholders for the development of clean energy systems (Ugwoke, Adeleke, Corgnati, Pearce, & Leone, 2020).

- The energy supply structure suggests the uptake of an integrated energy supply mix based on indigenous community RE sources. Biomass energy systems are not normally considered for rural electrification, but posits to be a more economic electrification option than diesel gensets which are usually suggested as backup systems (Rural Electrification Agency (REA) (2017b)). This affirms the need to delve into other readily accessible RE resources than focusing entirely on just solar energy (Ozoegwu, 2019). The development of RE based supply technologies can create more jobs per unit energy (Wei, Patadia, & Kammen, 2010) and generate income (Buchholz & Da Silva, 2010) than fossil fuel technologies because of their associated fuel and maintenance costs which produce economic flows within the community. Local policy makers could champion energy policies accordingly to realise these benefits, promote private investments and phase out or retrofit diesel plants to biomass plants. These policies could define targets for realizing the optimal technology mix at minimal system costs without compromising the security and resilience of the supply system. Also, localized policies should enable opportunities for faster local learning as it has a significant impact on the reducing costs of RE technology development (Huenteler, Niebuhr, & Schmidt, 2016).
- The future energy systems are encumbered by significant losses, therefore there is the need to drive policies and enforce regulations that require practitioners and utilities to curtail losses by inculcating more stringent supply side efficiency actions for power distribution. These actions could also foster social acceptance and be used to define soft policy instruments such as voluntary obligations (for example green pricing) which are non-binding with no direct impact on the market outcome (Savvidis et al., 2019), but can elicit goodwill from the citizens to adopt RE technologies and energy efficient behaviours.
- From the strategy's impact analysis (see Fig. 17), the results avail the opportunity to identify possible priorities and focal sectors across the different system elements and the composite scenarios. With limited funding impeding the development of RE-based energy systems (Karunathilake et al., 2018), this provides the opportunity to define strategic policy measures to pursue tiered or staggered electrification by focussing on particularly crucial sectors like the community sector and providing electricity access to health centres which has become the practice following the emergence of the global Covid-19 pandemic (Sustainable Energy for All, 2019).

The analyses also offer some new perspectives as regards tackling the Sustainable Development Goal 7 (SDG 7) and addressing the energy trilemma issues at local levels. The results seek to redress some misconceptions that has plagued rural energy planning across Africa (Moss, 2019). However, there are areas for improvement and future work including inclusion of an integrative scenario to assess the possibility of grid connection and associated cost. The implications for demand side management via efficient behaviours would also be considered by incorporating time-of-day pricing or other cost-reflective tariff mechanisms. Also, future work could determine the economic optimum as well as the social acceptability of the mini-grid system when interruptions are factored into the operation of the different power generation systems. The study by Schubert, Thuß, and Möst (2015) could provide the foundation for carrying out further investigations on the political and social feasibility dimensions on the development of these energy systems. Finally, the modelling framework used in the future could consider cooking energy use, which has been excluded as the necessary disaggregated data were not available at local scale.

5. Conclusion

C-SEP is a largely under-investigated scientific domain for the developing world together with exploring the synergies between CP and EP strategies. This study for the first time performed C-SEP from 2015 to 2040 for Onyen-okpon and Giere, two Nigerian rural communities. Utilizing the HI approach with the LEAP model, the implications for transforming rural communities to sustainable renewable based energy supply were investigated and the strategies synergies were quantitatively analysed based on four scenarios: the reference scenario and three composite scenarios. Overall, the composite scenarios reduced energy use, while achieving significant reductions in energy demand and GHGs emission through strategies promoting DSM and supply side efficiency.

The results show that the integration of strategies has clear benefits and the DSM3 scenario (based on advanced retrofit-IEQ_{III}) outperformed the other composite scenarios in terms of demand and emissions reductions, energy conservation, and overall reduction in energy expenditures. The integration approach provides useful insights for design and implementation of localised energy policies across different sectors in the community such that buttresses the need to incentivize productive energy uses. This avails the opportunity to concurrently define and reach multiple targets, goals and impacts. The insights gained would be applied to facilitate robust (scalable and modular) design of sustainable energy systems at local levels (integrable mini grids) across various time horizons (short, medium and long term). This would ensure sustainable exploitation of energy resources to deliver maximum utility in these

communities.

Nigeria has made good progress on developing regulatory frameworks geared towards mini-grid development (*Rural Electrification Agency (REA) (2017b)*). This calls for deliberate actions that are heavily reliant on detailed energy system analysis at local scales such as has been presented in this study. Therefore, programs like the Nigeria electrification project (*The World Bank, 2018*), championing the provision of solar hybrid mini grids for rural economic development would benefit from the findings of this analysis. Additionally, the methods and tools used here can be readily assessed and adopted through training of the local workforce to foster knowledge sharing and the implementation of the proposed strategies for drafting sustainable community energy master plans.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgements

This work has been facilitated by the financial support from Eni S.p.A Italy by virtue of the Eni Award 2017 prize and the Witte Endowment. Charlie Heaps and the Stockholm Environment Institute provided enough license time with LEAP. The anonymous reviewers are appreciated for their helpful comments.

References

- Aithal, S., & Rao, P. (2016). *Green education concepts & strategies in higher education model*. Retrieved January 2, 2021, from International Journal of Scientific Research and Modern Education (IJSRME) website: https://www.researchgate.net/publication/306031451_Green_Education_Concepts_Strategies_in_Higher_Education_Model.
- Alfaro, J. F., Miller, S., Johnson, J. X., & Riolo, R. R. (2017). Improving rural electricity system planning: An agent-based model for stakeholder engagement and decision making. *Energy Policy*, 101. <https://doi.org/10.1016/j.enpol.2016.10.020>
- Ali, T., Chiu, Y. R., Aghaloo, K., Nahian, A. J., & Ma, H. (2020). Prioritizing the existing power generation technologies in Bangladesh's clean energy scheme using a hybrid multi-criteria decision making model. *Journal of Cleaner Production*, 267. <https://doi.org/10.1016/j.jclepro.2020.121901>
- Allen, M. (2003). Liability for climate change. *Nature*. <https://doi.org/10.1038/421891a>
- Alstone, P., Gershenson, D., & Kammen, D. M. (2015). Decentralized energy systems for clean electricity access. *Nature Climate Change*, 5, 305–314. <https://doi.org/10.1038/nclimate2512>
- Bertok, B., & Bartos, A. (2020). Renewable energy storage and distribution scheduling for microgrids by exploiting recent developments in process network synthesis. *Journal of Cleaner Production*, 244. <https://doi.org/10.1016/j.jclepro.2019.118520>
- Bhattacharyya, S. C., & Timilsina, G. R. (2010). A review of energy system models. *International Journal of Energy Sector Management*, 4(4), 494–518. <https://doi.org/10.1108/17506221011092742>
- Buchholz, T., & Da Silva, I. (2010). Potential of distributed wood-based biopower systems serving basic electricity needs in rural Uganda. *Energy for Sustainable Development*, 14(1). <https://doi.org/10.1016/j.esd.2010.01.002>
- California, I. S. O. (2019). *Variable operations and maintenance cost definitions Report*. Retrieved from www.aiso.com.
- Chan, S. Y., & Chau, C. K. (2021). On the study of the effects of microclimate and park and surrounding building configuration on thermal comfort in urban parks. *Sustainable Cities and Society*, 64. <https://doi.org/10.1016/j.scs.2020.102512>
- Chen, H., & Chen, W. (2019). Carbon mitigation of China's building sector on city-level: Pathway and policy implications by a low-carbon province case study. *Journal of Cleaner Production*, 224. <https://doi.org/10.1016/j.jclepro.2018.12.022>
- Chen, H., Wang, Z., Xu, S., Zhao, Y., Cheng, Q., & Zhang, B. (2020). Energy demand, emission reduction and health co-benefits evaluated in transitional China in a 2 °C warming world. *Journal of Cleaner Production*, 264. <https://doi.org/10.1016/j.jclepro.2020.121773>
- Chen, Z., Huang, W., & Zheng, X. (2019). The decline in energy intensity: Does financial development matter? *Energy Policy*, 134, Article 110945. <https://doi.org/10.1016/j.enpol.2019.110945>
- Chen, B., Xiong, R., Li, H., Sun, Q., & Yang, J. (2019). Pathways for sustainable energy transition. *Journal of Cleaner Production*, 228. <https://doi.org/10.1016/j.jclepro.2019.04.372>
- Dai, H., Masui, T., Matsuoka, Y., & Fujimori, S. (2011). Assessment of China's climate commitment and non-fossil energy plan towards 2020 using hybrid AIM/CGE model. *Energy Policy*, 39(5), 2875–2887. <https://doi.org/10.1016/j.enpol.2011.02.062>
- de A. Colloço, F. M., Simoes, S. G., Dias, L. P., Duic, N., Seixas, J., & Bermann, C. (2019). The dawn of urban energy planning – Synergies between energy and urban planning for São Paulo (Brazil) megacity. *Journal of Cleaner Production*, 215. <https://doi.org/10.1016/j.jclepro.2019.01.013>
- Deichmann, U., Meisner, C., Murray, S., & Wheeler, D. (2011). The economics of renewable energy expansion in rural Sub-Saharan Africa. *Energy Policy*, 39(1). <https://doi.org/10.1016/j.enpol.2010.09.034>
- Dioha, M. O. (2017). Energy system models for Sub-Saharan African countries - a systematic review. *Journal of Sustainable Energy*, 8(4), 159–168. Retrieved from http://www.energy-cie.ro/archives/2017/nr_4/v8-n4-5.pdf.
- Dioha, M. O., & Emodi, N. V. (2019). Investigating the impacts of energy access scenarios in the Nigerian household sector by 2030. *Resources*, 8(3), 127. <https://doi.org/10.3390/resources8030127>
- Edmond, C. (2019). *Lake Chad shows how climate change is hitting real lives in Africa*. Retrieved December 4, 2019, from World Economic Forum website: <https://www.weforum.org/agenda/2019/12/lake-chad-africa-climate-change-water-indigenous-women/>.
- Edomah, N. (2019). Governing sustainable industrial energy use: Energy transitions in Nigeria's manufacturing sector. *Journal of Cleaner Production*, 210, 620–629. <https://doi.org/10.1016/j.jclepro.2018.11.052>
- Emodi, N. V., Emodi, C. C., Murthy, G. P., & Emodi, A. S. A. (2017). Energy policy for low carbon development in Nigeria: A LEAP model application. *Renewable and Sustainable Energy Reviews*, 68(September 2016), 247–261. <https://doi.org/10.1016/j.rser.2016.09.118>
- Energy Commission of Nigeria (ECN). (2014). *National energy master plan (Draft revised edition) 2014*. Retrieved from [http://www.energy.gov.ng/Energy_Policies_Plan/Draft\(Reviewed\)NEMP-2014.pdf](http://www.energy.gov.ng/Energy_Policies_Plan/Draft(Reviewed)NEMP-2014.pdf).
- ESMAP. (2007a). Technical and economic assessment of off-grid, mini-grid and grid electrification technologies. *ESMAP technical paper 121/07*. Retrieved from <http://bit.ly/1MKGBlh>.
- ESMAP. (2007b). *Technical and economic assessment of off-grid, mini-grid and grid electrification technologies*.
- Espinoza, A., Bautista, S., Narváez, P. C., Alfaro, M., & Camargo, M. (2017). Sustainability assessment to support governmental biodiesel policy in Colombia: A system dynamics model. *Journal of Cleaner Production*, 141. <https://doi.org/10.1016/j.jclepro.2016.09.168>
- Ezennaya, S. O., Isaac, O., Okolie, O. U., & Ezeanyim, O. C. (2014). Analysis of Nigeria's National Electricity Demand Forecast (2013–2030). *International Journal of Scientific & Technology Research*, 3(3), 333–340. Retrieved from www.ijstr.org.
- Farzaneh, H., Doll, C. N. H., & Puppim De Oliveira, J. A. (2016). An integrated supply-demand model for the optimization of energy flow in the urban system. *Journal of Cleaner Production*, 114. <https://doi.org/10.1016/j.jclepro.2015.05.098>
- Fuso Nerini, F., Dargaville, R., Howells, M., & Bazilian, M. (2015). Estimating the cost of energy access: The case of the village of Suro Craic in Timor Leste. *Energy*, 79(C), 385–397. <https://doi.org/10.1016/j.energy.2014.11.025>
- Gargiulo, M., Chiodi, A., De Miglio, R., Simoes, S., Long, G., Pollard, M., ... Giannakidis, G. (2017). An integrated planning framework for the development of sustainable and resilient cities - the case of the InSMART project. *Procedia Engineering*, 198. <https://doi.org/10.1016/j.proeng.2017.07.099>
- Geissler, S., Österreicher, D., & Macharm, E. (2018). Transition towards energy efficiency: Developing the Nigerian building energy efficiency code. *Sustainability*, 10(8), 2620. <https://doi.org/10.3390/su10082620>
- Geng, Y., Fujita, T., Bleischwitz, R., Chiu, A., & Sarkis, J. (2019). Accelerating the transition to equitable, sustainable, and livable cities: Toward post-fossil carbon societies. *Journal of Cleaner Production*, 239. <https://doi.org/10.1016/j.jclepro.2019.118020>
- Heaps, C. (2020). *LEAP: The low emissions analysis platform. [Software version: 2020.1.10]*. Somerville, MA, USA.
- Heidari, N., & Pearce, J. M. (2016). A review of greenhouse gas emission liabilities as the value of renewable energy for mitigating lawsuits for climate change related damages. *Renewable and Sustainable Energy Reviews*, 55. <https://doi.org/10.1016/j.rser.2015.11.025>
- Howells, M. I., Alfstad, T., Victor, D. G., Goldstein, G., & Remme, U. (2005). A model of household energy services in a low-income rural African village. *Energy Policy*, 33(14), 1833–1851. <https://doi.org/10.1016/j.enpol.2004.02.019>
- Huenteler, J., Niebuhr, C., & Schmidt, T. S. (2016). The effect of local and global learning on the cost of renewable energy in developing countries. *Journal of Cleaner Production*, 128. <https://doi.org/10.1016/j.jclepro.2014.06.056>
- Indra Al Irsyad, M., Halog, A. B., Nepal, R., & Koesrindartoto, D. P. (2017). Selecting tools for renewable energy analysis in developing countries: An expanded review. *Frontiers in Energy Research*, 5(December). <https://doi.org/10.3389/fenrg.2017.00034>
- Interagency Working Group on Social Cost of Greenhouse Gases, U. S. G. (2016). *Addendum to technical support document on social cost of carbon for regulatory impact analysis under executive order 12866: Application of the methodology to estimate the social cost of methane and the social cost of nitrous oxide*.
- IPCC. (2011). *Special report on renewable energy sources and climate change mitigation*.
- IRENA. (2012). *Renewable energy technologies: Cost analysis series- Biomass for Power Generation*. Retrieved from www.irena.org/Publications.
- IRENA. (2016). *Solar PV in Africa: Costs and markets*. Retrieved from www.irena.org.
- IRENA. (2018a). *Planning and prospects for renewable energy: West Africa*. Retrieved from http://www.irena.org/DocumentDownloads/Publications/IRENA_Africa_Power_Sector_synthesis_2015.pdf.
- IRENA. (2018b). *Renewable power generation costs in 2017*. International Renewable Energy Agency. https://doi.org/10.1007/SpringerReference_7300
- Jaccard, M. (2006). Sustainable fossil fuels: The unusual suspect in the quest for clean and enduring energy. *Sustainable fossil fuels: The unusual suspect in the quest for clean and enduring energy*. <https://doi.org/10.1017/CBO9780511754104>

- Jamshidi, M., & Askarzadeh, A. (2019). Techno-economic analysis and size optimization of an off-grid hybrid photovoltaic, fuel cell and diesel generator system. *Sustainable Cities and Society*, 44. <https://doi.org/10.1016/j.scs.2018.10.021>
- Jansen, S., Mohammadi, S., & Bokel, R. (2021). Developing a locally balanced energy system for an existing neighbourhood, using the 'Smart Urban Isle' approach. *Sustainable Cities and Society*, 64. <https://doi.org/10.1016/j.scs.2020.102496>
- Jones, E. C., & Leibowicz, B. D. (2021). Co-optimization and community: Maximizing the benefits of distributed electricity and water technologies. *Sustainable Cities and Society*, 64. <https://doi.org/10.1016/j.scs.2020.102515>
- Kadian, R., Dahiya, R. P., & Garg, H. P. (2007). Energy-related emissions and mitigation opportunities from the household sector in Delhi. *Energy Policy*, 35(12). <https://doi.org/10.1016/j.enpol.2007.07.014>
- Kanagawa, M., & Nakata, T. (2008). Assessment of access to electricity and the socio-economic impacts in rural areas of developing countries. *Energy Policy*, 36(6). <https://doi.org/10.1016/j.enpol.2008.01.041>
- Karunathilake, H., Perera, P., Ruparathna, R., Hewage, K., & Sadiq, R. (2018). Renewable energy integration into community energy systems: A case study of new urban residential development. *Journal of Cleaner Production*, 173. <https://doi.org/10.1016/j.jclepro.2016.10.067>
- Keirstead, J., Jennings, M., & Sivakumar, A. (2012). A review of urban energy system models: Approaches, challenges and opportunities. *Renewable and Sustainable Energy Reviews*, 16(6). <https://doi.org/10.1016/j.rser.2012.02.047>
- Li, L., Lei, Y., Wu, S., Huang, Z., Luo, J., Wang, Y., ... Yan, D. (2018). Evaluation of future energy consumption on PM2.5 emissions and public health economic loss in Beijing. *Journal of Cleaner Production*, 187. <https://doi.org/10.1016/j.jclepro.2018.03.229>
- Liu, Z., & Li, X. (2016). Power generation projects in China. *Advances in Social Science, Education and Humanities Research (ASSEHR). 2016 International Conference on Education, Management Science and Economics (ICEMSE-16)* (pp. 255–258). Atlantic Press, 65.
- Marten, A. L., & Newbold, S. C. (2012). Estimating the social cost of non-CO2 GHG emissions: Methane and nitrous oxide. *Energy Policy*, 51(Mc 1809), 957–972. <https://doi.org/10.1016/j.enpol.2012.09.073>
- Marzband, M., Alavi, H., Ghazimirsaeid, S. S., Uppal, H., & Fernando, T. (2017). Optimal energy management system based on stochastic approach for a home Microgrid with integrated responsive load demand and energy storage. *Sustainable Cities and Society*, 28. <https://doi.org/10.1016/j.scs.2016.09.017>
- Moore, F. C., & Diaz, D. B. (2015). Temperature impacts on economic growth warrant stringent mitigation policy. *Nature Climate Change*, 5. <https://doi.org/10.1038/nclimate2481>
- Moss, T. (2019). *Energy poverty: Why we are still failing Africa*. Retrieved January 5, 2020, from World Economic Forum website: <https://www.weforum.org/agenda/2019/11/energy-poverty-africa-sdg7/>
- Movahedian, Z., & Askarzadeh, A. (2018). Multi-objective optimization framework of a photovoltaic-diesel generator hybrid energy system considering operating reserve. *Sustainable Cities and Society*, 41. <https://doi.org/10.1016/j.scs.2018.05.002>
- Murphy, R., & Jaccard, M. (2011). Energy efficiency and the cost of GHG abatement: A comparison of bottom-up and hybrid models for the US. *Energy Policy*, 39(11). <https://doi.org/10.1016/j.enpol.2011.08.033>
- Nduka, E. (2021). How to get rural households out of energy poverty in Nigeria: A contingent valuation. *Energy Policy*, 149, Article 112072. <https://doi.org/10.1016/j.enpol.2020.112072>
- Neves, A. R., Leal, V., & Lourenço, J. C. (2015). A methodology for sustainable and inclusive local energy planning. *Sustainable Cities and Society*, 17. <https://doi.org/10.1016/j.scs.2015.04.005>
- Nigeria National Bureau of Statistics (NBS). (2019). *NBS E-library*. Retrieved February 26, 2019, from <https://www.nigerianstat.gov.ng/>
- Nwozor, A., Oshewolo, S., Owuoye, G., & Okidu, O. (2021). Nigeria's quest for alternative clean energy development: A cobweb of opportunities, pitfalls and multiple dilemmas. *Energy Policy*, 149, Article 112070. <https://doi.org/10.1016/j.enpol.2020.112070>
- Old School Value. (2020). *Practical guide to discount rates for value investors*. Retrieved March 12, 2020, from <https://www.oldschoolvalue.com/investing-strategy/explaining-discount-rates/>
- Ouedraogo, N. S. (2017). Modeling sustainable long-term electricity supply-demand in Africa. *Applied Energy*, 190, 1047–1067. <https://doi.org/10.1016/j.apenergy.2016.12.162>
- Ozoegwu, C. G. (2019). Artificial neural network forecast of monthly mean daily global solar radiation of selected locations based on time series and month number. *Journal of Cleaner Production*, 216, 1–13. <https://doi.org/10.1016/j.jclepro.2019.01.096>
- Pandey, R. (2002). Energy policy modelling: Agenda for developing countries. *Energy Policy*, 30(2), 97–106. [https://doi.org/10.1016/S0301-4215\(01\)00062-3](https://doi.org/10.1016/S0301-4215(01)00062-3)
- Pardo-García, N., Simoes, S. G., Dias, L., Sandgren, A., Suna, D., & Krook-Riekkola, A. (2019). Sustainable and Resource Efficient Cities platform – SureCity holistic simulation and optimization for smart cities. *Journal of Cleaner Production*, 215. <https://doi.org/10.1016/j.jclepro.2019.01.070>
- Pfenninger, S., Hirth, L., Schlecht, I., Schmid, E., Wiese, F., Brown, T., ... Wingenbach, C. (2018). Opening the black box of energy modelling: Strategies and lessons learned. *Energy Strategy Reviews*, 19. <https://doi.org/10.1016/j.esr.2017.12.002>
- Phdungsilp, A. (2010). Integrated energy and carbon modeling with a decision support system: Policy scenarios for low-carbon city development in Bangkok. *Energy Policy*, 38(9). <https://doi.org/10.1016/j.enpol.2009.10.026>
- Power for All. (2017). *Scaling up mini-grids through capital infusion, innovation, and cross-sector collaboration*. Retrieved December 4, 2019, from Medium website: <https://medium.com/energy-access-india/scaling-up-mini-grids-through-capital-infusion-innovation-and-cross-sector-collaboration-992c9b01ca7>
- REN 21. (2019). *Renewables global status report 2019* (Vol. 105). Renewables Now. Retrieved from https://www.ren21.net/wp-content/uploads/2019/05/gsr_2019_full_report_en.pdf
- Rocco, M. V., Tonini, F., Fumagalli, E. M., & Colombo, E. (2020). Electrification pathways for Tanzania: Implications for the economy and the environment. *Journal of Cleaner Production*, 263. <https://doi.org/10.1016/j.jclepro.2020.121278>
- Roche, M. Y., Ude, N., & Ofoegbu, I. D. (2017). *Comparison of costs of electricity generation in Nigeria*.
- Rogner, H. (2017). *Introduction to energy system modelling energy system*. Retrieved from <http://indico.ictp.it/event/8008/session/3/contribution/22/material/slides/0.pdf>
- Rojas-Zerpa, J. C., & Yusta, J. M. (2014). Methodologies, technologies and applications for electric supply planning in rural remote areas. *Energy for Sustainable Development*, 20(1), 66–76. <https://doi.org/10.1016/j.esd.2014.03.003>
- Rural Electrification Agency (REA). (2017a). *20 by 20: A design charrette to achieve 20cents/kWh by 2020*.
- Rural Electrification Agency (REA). (2017b). *The Off-grid Opportunity in Nigeria: Upscaling minigrids for least cost and timely access to electricity*. Retrieved from http://esmap.org/sites/default/files/Presentations/REA_Damila-Off-GridOpportunity_03122017_web.pdf
- Savvidis, G., Siala, K., Weissbart, C., Schmidt, L., Borggreffe, F., Kumar, S., ... Hufendiek, K. (2019). The gap between energy policy challenges and model capabilities. *Energy Policy*. <https://doi.org/10.1016/j.enpol.2018.10.033>
- Schubert, D. K. J., Thuß, S., & Möst, D. (2015). Does political and social feasibility matter in energy scenarios? *Energy Research and Social Science*, 7. <https://doi.org/10.1016/j.erss.2015.03.003>
- Sharifi, E., Larbi, M., Omran, H., & Boland, J. (2020). Climate change adaptation and carbon emissions in green urban spaces: Case study of Adelaide. *Journal of Cleaner Production*, 254. <https://doi.org/10.1016/j.jclepro.2020.120035>
- Shrestha, R. M., & Rajbhandari, S. (2010). Energy and environmental implications of carbon emission reduction targets: Case of Kathmandu Valley, Nepal. *Energy Policy*, 38(9). <https://doi.org/10.1016/j.enpol.2009.11.088>
- Sokona, Y., Mulugetta, Y., & Gujba, H. (2012). Widening energy access in Africa: Towards energy transition. *Energy Policy*, 47(1). <https://doi.org/10.1016/j.enpol.2012.03.040>
- Sola, A., Corchero, C., Salom, J., & Sanmarti, M. (2020). Multi-domain urban-scale energy modelling tools: A review. *Sustainable Cities and Society*, 54. <https://doi.org/10.1016/j.scs.2019.101872>
- St. Denis, G., & Parker, P. (2009). Community energy planning in Canada: The role of renewable energy. *Renewable and Sustainable Energy Reviews*, 13(8). <https://doi.org/10.1016/j.rser.2008.09.030>
- Sustainable Energy for All. (2019). *Lasting impact: Sustainable off-grid solar delivery models to power health and education*. Retrieved from <https://www.seforall.org/publications/lasting-impact-sustainable-off-grid-solar-delivery-models>
- Sustainable Energy for All. (2020). *Cooling as a Service: A disruptive business model for sustainable cooling*. Retrieved March 28, 2020, from News and stories website: <https://www.seforall.org/news/cooling-as-a-service-a-disruptive-business-model-for-sustainable-cooling>
- Szabó, S., Moner-Girona, M., Kougias, I., Bailis, R., & Bódis, K. (2016). Identification of advantageous electricity generation options in sub-Saharan Africa integrating existing resources. *Nature Energy*, 1(10), 1–8. <https://doi.org/10.1038/nenergy.2016.140>
- The World Bank. (2018). *Nigeria electrification project*. <https://doi.org/10.4324/9781315071299-37>
- Toman, M. A., & Jemelkova, B. (2003). Energy and economic development: An assessment of the state of knowledge. *The Energy Journal*, 4. <https://doi.org/10.5547/ISSN0195-6574-EJ-Vol24-No4-5>
- Torabi Moghadam, S., Delmastro, C., Corgnati, S. P., & Lombardi, P. (2017). Urban energy planning procedure for sustainable development in the built environment: A review of available spatial approaches. *Journal of Cleaner Production*, 165, 811–827. <https://doi.org/10.1016/j.jclepro.2017.07.142>
- Tudy, A. C. A. S. E. S., & Romising, O. F. A. P. (2017). *Mini-grids in Nigeria* (November).
- Twaha, S., & Ramli, M. A. M. (2018). A review of optimization approaches for hybrid distributed energy generation systems: Off-grid and grid-connected systems. *Sustainable Cities and Society*, 41. <https://doi.org/10.1016/j.scs.2018.05.027>
- Ugwoke, B., Adeleke, A., Corgnati, S. P., Pearce, J. M., & Leone, P. (2020). Decentralized renewable hybrid mini-grids for rural communities: Culmination of the IREP framework and scale up to urban communities. *Sustainability*, 12(18), 7411. <https://doi.org/10.3390/su12187411>
- Ugwoke, B., Becchio, C., Corgnati, S., Leone, P., & Pearce, J. M. (2020). *Adapting the European typology approach for building stock energy assessment (TABULA) for the developing world* (In review).
- Ugwoke, B., Gershon, O., Becchio, C., Corgnati, S. P., & Leone, P. (2020). A review of Nigerian energy access studies: The story told so far. *Renewable and Sustainable Energy Reviews*, 120, Article 109646. <https://doi.org/10.1016/j.rser.2019.109646>
- Ugwoke, B., Sulemanu, S., Corgnati, S., Leone, P., & Pearce, J. M. (2020). *Demonstration of the integrated rural energy planning framework for sustainable energy development in low-income countries: Case studies of rural communities in Nigeria* (In review).
- Van Beeck, N. M. J. P. (1999). Classification of energy models. In *Research memorandum* (Vol. 777). APA.
- Voigt, S., De Cian, E., Schymura, M., & Verdolini, E. (2014). Energy intensity developments in 40 major economies: Structural change or technology improvement? *Energy Economics*, 41. <https://doi.org/10.1016/j.eneco.2013.10.015>
- Walnum, H. T., Hauge, Å. L., Lindberg, K. B., Mysen, M., Nielsen, B. F., & Sornes, K. (2019). Developing a scenario calculator for smart energy communities in Norway: Identifying gaps between vision and practice. *Sustainable Cities and Society*, 46. <https://doi.org/10.1016/j.scs.2019.01.003>

- Wang, Y., Fang, X., Yin, S., & Chen, W. (2020). Low-carbon development quality of cities in China: Evaluation and obstacle analysis. *Sustainable Cities and Society*, 64. <https://doi.org/10.1016/j.scs.2020.102553>
- Wei, M., Patadia, S., & Kammen, D. M. (2010). Putting renewables and energy efficiency to work: How many jobs can the clean energy industry generate in the US? *Energy Policy*, 38(2). <https://doi.org/10.1016/j.enpol.2009.10.044>
- Weimer-Jehle, W., Buchgeister, J., Hauser, W., Kosow, H., Naegler, T., Poganietz, W. R., ... Vögele, S. (2016). Context scenarios and their usage for the construction of socio-technical energy scenarios. *Energy*, 111. <https://doi.org/10.1016/j.energy.2016.05.073>
- Wesseh, P. K., & Lin, B. (2017). Options for mitigating the adverse effects of fossil fuel subsidies removal in Ghana. *Journal of Cleaner Production*, 141. <https://doi.org/10.1016/j.jclepro.2016.09.214>
- Wienges, S., Bucher, A., & Hammer, S. A. (2016). *Nigeria (Intended) nationally determined contribution*. Retrieved from www.worldbank.org.
- Witt, T., Dumeier, M., & Geldermann, J. (2020). Combining scenario planning, energy system analysis, and multi-criteria analysis to develop and evaluate energy scenarios. *Journal of Cleaner Production*, 242. <https://doi.org/10.1016/j.jclepro.2019.118414>
- Witt, T., Stahlecker, K., & Geldermann, J. (2018). Morphological analysis of energy scenarios. *International Journal of Energy Sector Management*, 12(4), 525–546. <https://doi.org/10.1108/IJESM-09-2017-0003>
- Yadav, P., Davies, P. J., & Sarkodie, S. A. (2019). The prospects of decentralised solar energy home systems in rural communities: User experience, determinants, and impact of free solar power on the energy poverty cycle. *Energy Strategy Reviews*, 26, Article 100424. <https://doi.org/10.1016/j.esr.2019.100424>
- Yakubu, A., Ayandele, E., Sherwood, J., Olu, O. A., & Graber, S. (2018). *Minigrid investment report: Scaling the Nigerian market 2018*. Retrieved from https://rmi.org/wp-content/uploads/2018/08/RMI_Nigeria_Minigrid_Investment_Report_2018.pdf.
- Yılmaz Balaman, Ş., & Selim, H. (2016). Sustainable design of renewable energy supply chains integrated with district heating systems: A fuzzy optimization approach. *Journal of Cleaner Production*, 133. <https://doi.org/10.1016/j.jclepro.2016.06.001>
- Zafar, M. W., Shahbaz, M., Hou, F., & Sinha, A. (2019). From nonrenewable to renewable energy and its impact on economic growth: The role of research & development expenditures in Asia-Pacific Economic Cooperation countries. *Journal of Cleaner Production*, 212. <https://doi.org/10.1016/j.jclepro.2018.12.081>
- Zhou, L., & Chen, W. (2019). Impact of district-level decomposition policies to achieve a post-fossil carbon city: A case study of Beijing, China. *Journal of Cleaner Production*, 215. <https://doi.org/10.1016/j.jclepro.2019.01.082>